Assessing species status and genetic diversity in the Tombigbee Darter (*Etheostoma lachneri*)

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The Tombigbee Darter (*Etheostoma lachneri*) is the sole member of the snubnose subgenus *Ulocentra* endemic to the Tombigbee River Watershed of Mississippi and Alabama. *Etheostoma lachneri* is a species of greatest conservation need, due to suspected population decline since 1994 (Suttkus et al., 1994). To assess the persistence and genetic variability of *E. lachneri*, a population survey at 40 historical sites in Mississippi was performed. Results showed that 12 out of the 40 sites did not have *E. lachneri* present. Genomic analyses were conducted for populations across the species range using ddRADseq, revealing the possible complex of three species. To assess environmental DNA (eDNA) detection, *E. lachneri* mitogenome was sequenced. Results showed that *E. lachneri* eDNA was present at 10 of 39 survey sites but was less accurate than traditional sampling. Results presented here will contribute to the prioritization of *E. lachneri* in the Mississippi and Alabama statewide action plans.

DEDICATION

To my parents who made me love the outdoors and inspire me every day. To my siblings, Madison and Andrew, for always supporting me and motivating me to obtain my master's degree. I would also like to thank my former mentor Dr. Mark Meade for getting me into ichthyology and teaching me about all the fun critters living in our creeks and streams.

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CHAPTER I

POPULATION ASSESSMENT OF THE TOMBIGBEE DARTER (ETHEOSTOMA LACHNERI) THROUGH HISTORICAL TENNESSEE TOMBIGBEE WATERSHED SITES IN MISSISSIPPI

1.1 Background

Etheostoma lachneri (Tombigbee Darter) is a snubnose darter in the subgenus *Ulocentra* in the family Percidae. *Etheostoma lachneri* was described by Suttkus and colleagues (1994), but historical records began describing the species as *Etheostoma* (*Ulocentra*) sp. As early as 1961 (Mettee et al., 1989; Suttkus et al., 1994). Male *E. lachneri* exhibit four to six green-blue bars across the side of the body interspersed by orange-red blotches. Additionally, there are eight saddle blotches across the anterior dorsal surface. Dark green, red, and olive-green coloration is present on the spiny dorsal and soft-rayed dorsal fins. Females do not exhibit bright coloration and have a light brown color with dark green blotches across the side of the body. After the breeding season, males no longer display dimorphic coloration, appearing similar to females (Suttkus et al., 1994). While it is unknown which invertebrates comprise *E. lachneri* diets, other darters in the subgenus *Ulocentra,* such as *Etheosoma chermocki* and *Etheostoma duryi*, have diets composed mostly of small benthic invertebrates and their larvae (Khudamrongsawat et al., 2005; Mettee et al., 1989). Although little habitat preference data is available for *E. lachneri*, other snubnose darter species require pools, runs, gravel/rocky debris, and submerged vegetation to survive and reproduce (Ross, 2001). Some snubnose darters such as *Etheostoma chermocki*

require clear, shallow streams with a slow to medium current for survival (Khudamrongsawat et al., 2005). During the spawning season, early spring to summer, snubnose darters use gravel and small rock substrate to lay their eggs and conceal them from predators (Porterfield, 1998). *Etheostoma lachneri* is currently listed as vulnerable in Mississippi but listed as secure for Alabama (Hammerson & Clausen, 2011). Population surveys have been conducted in 1994, 1996, and 2020 (Kreiser & Sterling, 2020; Mettee et al., 1989; Ross, 2001; Suttkus et al., 1994). While some data exists for *E. lachneri*, there is limited data on the current occurrence of the species in Mississippi.

1.1.1 Distribution

In the Geological Survey of Alabama Atlas 24 titled "*Fishes of the Western Mobile River Basin in Alabama and Mississippi*", populations of *Etheostoma* sp. are listed, which is believed to be what is known today as *E. lachneri* (Mettee et al., 1989). Distribution is shown throughout the western Mobile River basin covering Appalachian Plateaus, Valley and Ridge, and Coastal Plain Province. Historical and current occupancy data indicate there are fewer occurrences in the Black Belt region.

1.1.2 Phylogeny

Darters are one of the most diverse groups of fish in the southeastern United States, with over 250 species (Near et al., 2011). Recent literature indicates a common trend of speciation occurring within the subgenus *Ulocentra* (Kim et al., 2023; Kozal et al., 2017; Kreiser & Sterling, 2020). In 2011, Near et al. constructed one of the first phylogenetic trees of the subgenus *Ulocentra*, which is now composed of 26 species and six undescribed species (Kim et al., 2023; Kozal et al., 2017; Near et al., 2011; Sterling & Warren, 2020; Table 1.1). Of the

twenty-six species, seven are listed as imperiled and two as critically imperiled in the *NatureServe Explorer* database (Hammerson & Clausen, 2011). Over 34% of the subgenus have declining populations, and six have no data on their population status.

1.1.3 Conservation concerns

Snubnose darters are increasingly threatened by anthropogenic disturbances, such as climate change and habitat loss (Sterling & Warren, 2020). Climate change is causing fluctuations in temperature, pH, and average flow rate in streams, which may result in imperilment of more snubnose darters. While darters can acclimate to environmental changes seasonally, the rapid pace of climate change may limit their ability to adapt (Firth et al., 2024). As water temperatures rise, dissolved oxygen decreases, causing fishes to increase their ventilation rates in an effort to extract more oxygen (Kramer, 1987). This increased water ventilation demands more energy from the fish. Rising temperatures can impair the ion exchange in darter gills, leading to a decrease in overall fitness (Firth et al., 2024). As summer temperatures increase, darters may become less tolerant of elevated water temperatures (Hlohowskyj & Wissing, 1987). Furthermore, snubnose darters require high levels of oxygen, particularly in warmer stream conditions, which makes their survival in a warmer climate even more challenging (Ultsch et al., 1978).

Alongside challenges imposed by water temperature, changes in hydrology also affect darter fitness. Most streams have culverts, which can further affect the fitness of small-bodied fish such as darters by preventing passage. Habitat loss is another factor that has affected these darter populations (Smith & Fausch, 1997). Most river and stream systems in the United States have been dredged for channelization, which alters flow rate (Brooker, 1985). When dredging occurs, the bottom substrate of a body of water is altered and typically changes from rocky to silt

(Pennington & Baker, 1982). Additionally, dredging can release toxic gases from deep substrates, which can be harmful to fish health (Yan & Li, 2023). Further research needs to be conducted to determine not only the habitats these darters occupy, but also to assess the current condition of these areas in relation to the future ecological success of these species. With the current lack of population data for *E. lachneri*, the effect of anthropogenic interference is relatively undocumented. Thus, further investigation into population estimates is required.

1.1.4 Physiology

Due to the conservation concerns and the physiology of *E. lachneri*, the species may be unable to adapt to anthropogenic disturbances. Darters in the subgenus *Ulocentra* inhabiting shallow streams are subjected to more dramatic fluctuations in water temperature, which could negatively impact species less tolerant to rapid temperature fluctuations (Johnston & Maceina, 2009). Furthermore, *E. lachneri* is a benthic species that lacks swim bladders, meaning they must expend more energy to reach the surface of the water (Evans & Page, 2003). As an adaptation of occupying fast-moving streams, they have round bodies and large fins compared to their body size (Evans & Page, 2003). Due to mainly swimming in short bursts, darters are confined by their inability to swim for long durations. Since darters cannot swim for extended periods, relocating to more favorable stream conditions is challenging. Due to these evolutionary constraints, the physiology of some darter species has become highly specialized to specific habitat features (Ultsch et al., 1978). While phenotypic variation in fish populations will allow some individuals to survive, the species is expected to decrease due to unfavorable habitat from anthropogenic disturbances (Ficke et al., 2007).

1.1.5 Tennessee Tombigbee Waterway

The southeastern United States is recognized as an aquatic biodiversity hotspot, with many species endemic to the region (Elkins et al., 2019). Despite this high biodiversity, many native species are facing imperilment due to anthropogenic disturbances. One such disturbance is the Tennessee Tombigbee Waterway (TTW), the man-made canal connecting the Tennessee and Tombigbee Rivers. The TTW was built to allow quicker barge routes to the Gulf of Mexico. The waterway's construction began in the early 1970's despite pushback from the public and U.S. Congress regarding the expense and environmental impacts (Stine, 1991). Construction finished in 1984. As a result of dredging, the bottom substrate changed from sandy/gravel to sandy/silt (Pennington & Baker, 1982). There are 10 locks and dams along the TTW, which present many threats to migratory fish species and threaten aspects of the stream itself, such as flow regimes, physical factors including stream substrate, and chemical factors including nutrient cycling (Oliveira et al., 2018). Connectivity of the Tennessee River to the Tombigbee River caused by the TTW has allowed the spread of non-native taxa into these areas. This has increased competition for resources among fish, potentially decreasing native populations (Mills et al., 2004). Studies have shown that culverts and dams can cause habitat fragmentation to occur (Lehrter et al., 2024; Zarri et al., 2022). Due to these findings and the multiple dams and culverts on the TTW and tributaries, *E. lachneri* populations could lead to genetic variation among isolated populations over time. With a lack of data, funding, and resources, the status of many darter species residing in TTW tributaries, including *E. lachneri,* remains unknown. Furthermore, while overall genetic variation benefits the species, long term isolation can lead to inbreeding, creating a bottleneck effect and may eventually result in extirpation (Hänfling &

Brandl, 1998). Due to anthropogenic disturbances such as the construction of the TTW and a gap in current occupancy data, *E. lachneri* could be experiencing local extirpation.

1.1.6 Conservation efforts

Etheostoma lachneri was listed in the Mississippi 2015 State Wildlife Action Plan (Mississippi Museum of Natural Science, 2015). The action plan categorized the darter as a tier 2 species, indicating the need for conservation action. It was given a global rank of 4, indicating that the darter is uncommon but not rare; however, there are concerns for its long-term status (IUCN, 2025). In Mississippi, the species is ranked S3, indicating it is vulnerable due to rarity, limited range, or other factors that could lead to its extirpation (IUCN, 2025). The Mississippi Action Plan states that there are knowledge gaps regarding habitat structure and distribution across the state. While there is still a high need for understanding the species reproduction and habitat requirements, our research aimed to survey historical and potential sites that the darter has inhabited or could reside.

This chapter uses standardized electrofishing sampling methods in 40 wadeable streams in Mississippi TTW tributaries to evaluate if populations of *E. lachneri* are secure. Alternatively, due to anthropogenic disturbances such as the construction of the TTW, *E. lachneri* could be exhibiting local extirpation. The results of this study will improve understanding of this species of greatest conservation need, under the Mississippi State Wildlife Action Plan (Mississippi Museum of Natural Science, 2015).

1.2 Methods

1.2.1 Site selection

Using historical data from the Mississippi Department of Wildlife, Fisheries and Parks (MDWFP) and FishMap.org database (http://fishmap.org/index.html) for *E. lachneri* (Crutchfield, 2013), 40 sites were selected where *E. lachneri* has been observed historically. Hydrologic Unit Codes (HUC) are a numbering unit for surface hydrological watersheds. The HUC 8 localities are the best spatial scale to analyze genetic diversity (Zbinden et al., 2023). Surveys were conducted in the following HUC 8 units: Sucarnoochee (HUC ID:03160202), Noxubee (HUC ID:03160108), Middle Tombigbee Lubbub (HUC ID:03160106), Tibbee (HUC ID:03160104), Luxapalila (HUC ID:03160105), Buttahatchee (HUC ID:03160103), Upper Tombigbee (HUC ID:03160101), and Town (HUC ID: 03160102). If multiple sites were selected within a HUC 8, priority was given to sites on different drainages. The HUC 8 localities from the Upper-Tombigbee HUC had the most historical observations.

1.2.2 Survey Methods

Streams were measured in four 30-meter transects. Each transect was allotted 15 minutes for the capture of fish species. Time was paused after each electrofishing effort to identify and count any ichthyofauna caught. Our team used a Smith-Root® LR-24 backpack electrofisher (Smith-Root, Vancouver, Washington), dip nets, and a 6x4 or 6x10 m seine. All stream types, pools, riffles, and runs were accounted for during the survey. To account for potential bias, each survey was conducted by four to five surveyors. Streams that were too deep for backpack electrofishing were replaced with a shallower channel at the same site. Fish and other aquatic species were identified, counted, and recorded to evaluate occupancy of the surveyed streams. This helped determine whether there was a statistical relationship between *E. lachneri*

occurrence and specific species observed in these streams. Catch per unit effort (CPUE) was calculated for *E. lachneri* by dividing the number of individuals caught at each site by the total time of the survey.

1.2.3 Statistical Methods

To illustrate relationships between fish community data and *E. lachneri,* a neighbor-joining clustering analysis was performed in the software PAST (v.4.30) by comparing fish community presence data with each site. The neighbor-joining clustering analysis was re-run without *E. lachneri* in the data set, then the two output files, with and without *E. lachneri* were combined and visualized using a tanglegram in Dendroscope v.3.8.10 (Hammer et al., 2001). A principal component analysis (PCA) was performed in PAST to infer relationships between species present throughout the 40 sites surveyed using binary data. An indicator species analysis was performed in R™ (R Core Team, 2024) in the RStudio™ interface (v.2024.04.2) using the *multipatt* function in the *indicspecies* package to examine which fish species correlate with sites surveyed. This matrix is on a scale of 0 to 1, with 1 resulting in high correlation. Species richness index was estimated in R™ (R Core Team, 2024) using the *estimateR* function in the *vegan* package, to see which sites had greater species richness. This analysis indicated the probability that other species were present at the site that may not have been collected during the survey.

A Logistic Regression Model was conducted using the command *glm* with the package *stats* to show if a site's distance from the TTW predicted with the presence of *E. lachneri* (R Core Team, 2024). The site's distance from the TTW was calculated using the near tool in ArcGIS Pro. Additionally, to see if there is a relationship between a site's CPUE for *E. lachneri* and a site's distance from the TTW, a Generalized Linear Model-Poisson was run using the command *glm* with the package *stats* (R Core Team, 2024).

1.3 Results

Survey results showed that *E. lachneri* were observed at 70% of sites (Table 1.2; Figure 1.1). Across all sites, the survey yielded 10,758 individuals, including 88 species (Table A.1). The most abundant species was *Notropis baileyi*. Indicator species analysis indicated that sampled sites had a high correlation with *Gambusia affinis* (Permutation stat=0.993, p=0.003), *Nocomis leptocephalus* (stat=0.874, p=0.124), *Percina nigrofasciata* (stat=0.874, p=0.095), *Notropis ammophilus* (stat=0.0.863, p=0.135), and our target species, *E. lachneri* (stat=0.0.858, p=0.155; Table 1.3).

The CPUE for *E. lachneri*, had the highest rating at the following sites: Gum Creek (KMF23081402), Johns Creek (KMF23080303), and Wolf Creek (DLP23082601) in HUC-8 Upper-Tombigbee (Table 1.4). For the species richness index, sites had a total of 78 taxa, with a Chao1 value of 88.97, a jackknife species richness suggesting an even greater richness (93-97 species), and a standard error of 7.42 (Table 1.5). The tanglegram from the two combined Neighbor Joining cluster analyses, one with and one without *E. lachneri*, shows that community relationships change when *E. lachneri* was present for the following species: *Fundulus notatus, Hypentelium etowanum, Cyprinella venusta,* and *Etheostoma artesia* (Figure 1.2). Principal component analysis based on correlation data showed *E. lachneri* occupied similar areas as *Nocomis leptocephalus*, *Notropis baileyi*, and *Percina nigrofasciata*, indicating that all have similar collection data based on presence data in addition to similar habitat needs (Figure 1.3; Ross, 2001). The Logistic Regression Model results showed that the site's distance to the TTW does not predict presence of *E. lachneri* (Z= -0.420, *P*=0.6747). Moreover, results from a Generalized Linear Model-Poisson showed that a site's distance to the TTW did not influence the CPUE of *E. lachneri* (beta estimate: -0.098, *P*=0.734).

1.4 Discussion

This study was able to evaluate *E. lachneri* populations and sites' CPUE values. Regression models could not detect if *E. lachneri* populations were affected by the construction of the TTW and if populations are currently secure. A lack of observations at sites does not equate to a lack of habitat occupation (Gu & Swihart, 2004). In addition, limitations to sampling sites with road access could have skewed our results, due to the presence of bridges or culverts. Installation of bridges and culverts is known to disturb substrate and aquatic vegetation, which can ultimately disrupt *E. lachneri* habitats (Suttkus et al., 1994). It is additionally important to note that observed densities of *E. lachneri* are not an indicator of stream quality, on account of confinement as a result of physical and ecological barriers (Van Horne, 1983).

Our regression models (logistic and Poisson glm) failed to detect a relationship between a site's distance to the TTW to the presence of *E. lachneri*, nor their CPUE value. While statistical analysis based on geographic distance cannot determine the TTW's effect on *E. lachneri*, heavy silt deposits were present at each site. Previous research has shown that human activities, such as dredging rivers and streams, have led to a decline in fish populations (Wenger et al., 2018). Even though heavy silt deposits were present at each site, this study cannot assert that the construction of the TTW has affected the *E. lachneri* population due to insufficient historical data on each site's substrate prior to the building of the TTW.

Indicator species analysis showed that the species most correlated with our study sites had similar occurrences throughout the southeast US (Ross, 2001). Given that the PCA of species presence at each of the 40 sites surveyed showed that *E. lachneri*, *Noturus leptacanthus, Notropis baleyi,* and *Percina nigrofasciata* all had similar occurrence values, they are likely to have similar realized niches. While all four species seem to have similar habitats, *E. lachneri*

populations are not as widespread throughout Mississippi. This could be due to other darters in the subgenus *Ulocentra* being present such as *Etheostoma histrio, Etheostoma raneyi,* and *Etheostoma zonistium*. Studies have shown that in some areas, specific darter species are replaced with another darter species due to available habitat and food resources, or they can outcompete other darters by having a larger body size (Hubbell et al., 2020). The tanglegram of the Neighbor Joining Analyses supports the argument that *E. lachneri* has an effect on the local fish community, and without *E. lachneri* present, there could be a decrease in species composition. Since this survey only occurred in Mississippi, populations in Alabama need to also be assessed.

Species richness estimates show that there were about 11 more species that were not observed. Due to these findings, there is support for *E. lachneri* potentially being present at sites where the species was not detected during the sampling events. While habitat data was not collected for this study, greater diversity ratings for some of these HUCs could be due to less anthropogenic disturbances in the stream such as less agricultural runoff.

Overall, standardized sampling methods reveal that *E. lachneri* seems to be stable and distance from the TTW did not affect its CPUE nor occupancy. While results show that *E. lachneri* is stable, sequencing genetics will help identify which populations require more management and could explain *E. lachneri* range distribution.

1.5 Tables

Table 1.1 A comprehensive list of all known snubnose darters along with their conservation status.

Snubnose Darters	**Conservation Status**
Etheostoma baileyi	Apparently Secure
Etheostoma histrio	Secure
Etheostoma scotti	Imperiled
Etheostoma coosae	Apparently Secure
Etheostoma duryi	Apparently Secure
Etheostoma flavum	Apparently Secure
Etheostoma etnieri	Apparently Secure
Etheostoma raneyi	Imperiled
Etheostoma faulkneri	Imperiled
Etheostoma cyanoprosopum	Critically Imperiled
Etheostoma zonistium	Apparently Secure
Etheostoma zonistium Hatchee	Unknown
Etheostoma cervus	Imperiled
Etheostoma pyrrhogaster	Imperiled
Etheostoma brevirostrum	Imperiled
Etheostoma ramseyi	Apparently Secure
Etheostoma tallapoosae	Apparently Secure
Etheostoma colorosum	Secure
Etheostoma lachneri	Apparently Secure
Etheostoma cf. bellator (Sipsey darter)	*Unknown*
Etheostoma cf. Bellator (Locust Fork Darter)	*Unknown*
Etheostoma cf. bellator -Valley Creek	*Unknown*
Etheostoma bellator	Imperiled
Etheostoma cf. bellator -Gurley Creek	Unknown
Etheostoma chermocki	Critically Imperiled
Etheostoma cf. bellator -Little Warrior River	Unknown

Notes: Data from Near et al., 2011; Kozal et al., 2017; Sterling & Warren, 2020; Kim et al., 2023; and Hammerson & Clausen, 2011.

Table 1.2 Site location and date surveyed for 40 standardized aquatic faunal surveys.

Site ID	Date	Latitude	Longitude	Site Name
MWS23062301	6/23/2023	32.9938	-88.6545	Running Water Creek
MWS23062302	6/23/2023	33.1223	-88.8063	Sun Creek
MWS23062701	6/27/2023	33.1975	-88.9947	Mill Creek
KMF23071101	7/11/2023	33.8304	-88.9437	Chico Creek
KMF23071102	7/11/2023	34.0870	-88.9913	Redland Creek
KMF23071801	7/18/2023	34.5194	-88.4455	Hurricane Creek
KMF23071802	7/19/2023	34.6677	-88.4294	Big Brown Creek
KMF23072601	7/26/2023	33.5088	-88.3932	Luxapalila Creek
KMF23072602	7/26/2023	33.4776	-88.3628	McCary Creek
KMF23072603	7/26/2023	33.4800	-88.3602	Catalpa-Tibbee Creek
KMF23072801	7/28/2023	33.0447	-88.8172	Moody Creek
KMF23072802	7/28/2023	32.8356	-88.6588	Straight Creek
KMF23073101	7/31/2023	33.8323	-88.4749	Nichols Creek
KMF23073102	7/31/2023	33.8559	-88.3194	unnamed Creek-Buttahatchee River
KMF23073103	7/31/2023	33.7577	-88.4184	Wilson Creek
KMF23080301	8/3/2023	34.0899	-88.3497	Bluff Creek
KMF23080302	8/3/2023	34.1185	-88.3157	Smith Creek
KMF23080303	8/3/2023	34.2303	-88.2577	Johns Creek
KMF23081101	8/11/2023	32.6896	-88.6164	Pawticfaw Creek
KMF23081102	8/11/2023	32.6223	-88.6212	Blackwater Creek
KMF23081401	8/14/2023	34.2859	-88.2188	Briar Creek
KMF23081402	8/14/2023	34.3850	-88.2056	Gum Creek
KMF23081403	8/14/2023	34.3045	-88.3546	Dulaney Branch
KMF23081501	8/15/2023	34.4718	-88.2609	Red Bud Creek
KMF23081502	8/15/2023	34.6682	-88.2846	Hurricane Creek
DLP23081901	8/19/2023	32.5298	-88.5118	Ponta Creek
DLP23081902	8/19/2023	32.3659	-88.4153	Alamuchee Creek
DLP23082601	N/A	33.8680	-88.5687	Wolf Creek
DLP23082602	N/A	33.6653	-88.2796	Mayhew Creek
KMF23090401	9/4/2023	33.9441	-88.4875	Weaver Creek
KMF23090402	9/4/2023	34.1636	-88.5106	Boguefala Creek
KMF23090403	9/4/2023	34.2297	-88.6667	Tulip Creek
KMF23090901	9/9/2023	34.5837	-88.6276	Twenty Mile Creek
KMF23090902	9/9/2023	34.3093	-88.5101	Mantachie Creek
KMF23090903	9/9/2023	34.4847	-88.5060	Donivan Creek
KMF23091301	9/13/2023	33.3822	-88.3130	Nash Creek
KMF23091302	9/13/2023	33.4094	-88.3336	Ellis Creek

Table 1.2 (continued)

Site ID	Date	Latitude	Longitude	Site Name
KMF23091303	9/13/2023	33.5353	-88.3040	Brush Creek
KMF23092301	9/23/2023	34.1099	-88.7222	Chiwapa Creek
KMF23092302	9/23/2023	34.2208	-88.9994	Webster Creek

Table 1.3 Indicator species analysis based on 40 standardized aquatic faunal surveys.

Species	s.1	s.2	index	stat	p.value
Gambusia affinis	1	0	1	0.993	0.003
Nocomis leptocephalus	0	1	2	0.874	0.124
Percina nigrofasciata	0	1	2	0.874	0.095
Notropis ammophilus	1	0	1	0.863	0.135
Etheostoma lachneri	0	1	2	0.858	0.155
Lepomis solis	1	1	3	0.851	NA
Lythrurus bellas	1	1	3	0.837	NA
Notropis baileyi	0	1	2	0.827	0.245
Lepomis macrochirus	1	1	3	0.822	NA
Luxilus chrysocephalus	0	1	2	0.811	0.284
Micropterus henshalli	0	1	2	0.811	0.284
Lepomis cyanellus	1	1	3	0.806	NA
Semotilus atromaculatus	1	1	3	0.791	NA
Noturus leptacanthus	0	1	2	0.743	0.374
Campostoma anomalum	0	1	2	0.707	0.405
Etheostoma nigrum	0	1	2	0.707	0.379
Noturus nocturnus	0	1	2	0.707	0.419
Lepomis microlophus	1	0	1	0.699	0.062
Cyprinella venusta	1	0	1	0.698	0.116
Micropterus salmoides	1	0	1	0.694	0.059
Pimephales notatus	1	1	3	0.689	NA
Fundulus olivaceous	0	1	2	0.688	0.467
Etheostoma rupestre	0	1	2	0.669	0.441
Etheostoma stigmaeum	0	1	2	0.669	0.428
Pimephales vigilax	1	0	1	0.669	0.081
Etheostoma swainii	0	1	2	0.649	0.511
Notropis texanus	0	1	2	0.628	0.511
Etheostoma artesiae	1	1	3	0.592	NA

Table 1.3 (continued)

Species	**s.1**	**s.2**	**index**	**stat**	**p.value**
Lamprey spp.	0	1	2	0.585	0.664
Moxostoma poecilurum	0	1	2	0.562	0.735
Fundulus notatus	1	1	3	0.548	NA
Etheostoma parvipinne	1	0	1	0.529	0.242
Hypentelium etowanum	0	1	2	0.487	0.937
Percina sciera	1	1	3	0.474	NA
Elassoma zonatum	0	1	2	0.459	1.000
Erimyzon oblongus	0	1	2	0.429	1.000
Noturus gyrinus	0	1	2	0.429	1.000
Lepomis gulosus	0	1	2	0.363	1.000
Lepomis megalotis	0	1	2	0.363	0.856
Semotilus thoreauianus	0	1	2	0.363	1.000
Amieurus natalis	0	1	2	0.324	1.000
Etheostoma chlorosomum	0	1	2	0.324	1.000
Etheostoma proeliare	0	1	2	0.324	1.000
Ictiobus punctatus	0	1	2	0.324	1.000
Lepomis miniatus	0	1	2	0.324	1.000
Notropis spp.	0	1	2	0.324	1.000
Ambloplites ariommus	0	1	2	0.281	1.000
Aphredoderus sayanus	0	1	2	0.281	1.000
Esox americanus	0	1	2	0.281	1.000
Etheostoma spp.	0	1	2	0.281	1.000
Hybognathus nucholi	0	1	2	0.281	1.000
Hybopsis wincelli	0	1	2	0.281	1.000
Notropis volucellus	0	1	2	0.281	0.944
Ammocrypta beanii	0	1	2	0.229	1.000
Cambarus striatus	0	1	2	0.229	1.000
Labidesthes vanhyningi	0	1	2	0.229	1.000
Macrhybopsis	0	1	2	0.229	1.000
Minytrema melanops	0	1	2	0.229	1.000
Notropis buccata	0	1	2	0.229	1.000
Notropis candidus	0	1	2	0.229	1.000
Percina kathae	0	1	2	0.229	1.000
Percina maculata	0	1	2	0.229	1.000
Percina vigil	0	1	2	0.229	1.000
Ammocrypta meridiana	0	1	2	0.162	1.000
Crystallaria asprella	0	1	2	0.162	1.000

Table 1.3 continued

Species	s.1	s.2	index	stat	p.value
Erimyzon claviformes	0	1	2	0.162	1.000
Etheostoma fusiforme	0	1	2	0.162	1.000
Ictalurus bubalus	0	1	2	0.162	1.000
Lepomis marginatus	0	1	2	0.162	1.000
Lepomis punctatus	0	1	2	0.162	1.000
Micropterus spp.	0	1	2	0.162	1.000
Moxostoma spp.	0	1	2	0.162	1.000
Notropis atherinoides	0	1	2	0.162	1.000
Noturus funebris	0	1	2	0.162	1.000
Noturus munitus	0	1	2	0.162	1.000
Noturus phaeus	0	1	2	0.162	1.000
Percina spp.	0	1	2	0.162	1.000
Percina suttkusi	0	1	2	0.162	1.000
Carpiodes cyprinus	NA	NA	NA	NA	NA
Ichthyomyzon gagei	NA	NA	NA	NA	NA
Notemigonus crysoleucas	NA	NA	NA	NA	NA

Abbreviations: Species associated with group 1 (s.1), species associated with group 2 (s.2) indicator value index (index), permutation test association statistic (stat), and significance of the association (p.value).

Table 1.4 Catch Per Unit Effort (CPUE) data (number of *E. lachneri* individuals caught per hour) collected from 40 sampling sites using electrofishing, including stream locality and site field number.

Field Number	Creek Locality	CPUE
MWS23062301	Running Water Creek	0
MWS23062302	Sun Creek	0
MWS23062701	Mill Creek	9
KMF23071101	Chico Creek	0
KMF23071102	Redland Creek	22
KMF23071801	Hurricane Creek	2
KMF23071802	Big Brown Creek	0
KMF23072601	Luxapalila Creek	0
KMF23072602	McCary Creek	0
KMF23072603	Catalpa-Tibbee Creek	0
KMF23072801	Moody Creek	1

Table 1.4 (continued)

Field Number	Creek Locality	CPUE
KMF23072802	Straight Creek	7
KMF23073101	Nichols Creek	2
KMF23073102	unnamed Creek-Buttahatchee River	0
KMF23073103	Wilson Creek	8
KMF23080301	Bluff Creek	5
KMF23080302	Smith Creek	5
KMF23080303	Johns Creek	44
KMF23081101	Pawticfaw Creek	16
KMF23081102	Blackwater Creek	33
KMF23081401	Briar Creek	27
KMF23081402	Gum Creek	33
KMF23081403	Dulaney Branch	4
KMF23081501	Red Bud Creek	8
KMF23081502	Hurricane Creek	0
DLP23081901	Ponta Creek	8
DLP23081902	Alamuchee Creek	14
DLP23082601	Wolf Creek	53
DLP23082602	Mayhew Creek	8
KMF23090401	Weaver Creek	15
KMF23090402	Boguefala Creek	11
KMF23090403	Tulip Creek	14
KMF23090901	Twenty Mile Creek	0
KMF23090902	Mantachie Creek	18
KMF23090903	Donivan Creek	6
KMF23091301	Nash Creek	3
KMF23091302	Ellis Creek	5
KMF23091303	Brush Creek	3
KMF23092301	Chiwapa Creek	0
KMF23092302	Webster Creek	0

Table 1.5 Species richness index average for 40 standardized aquatic faunal surveys.

Species	**chao**	**chao.se**	**jack1**	**jack1.se**	**jack2**	**boot**	**boot.se**	**n**
78	88.96875	7.421839	92.625	7.430301	97.61859	85.17173	4.342814	40

Notes: Chao refers to the estimate number of species in the community. Jack1 refers to the first order Jackknife estimation. Boot refers to bootstrap or resampling method.

1.6 Figures

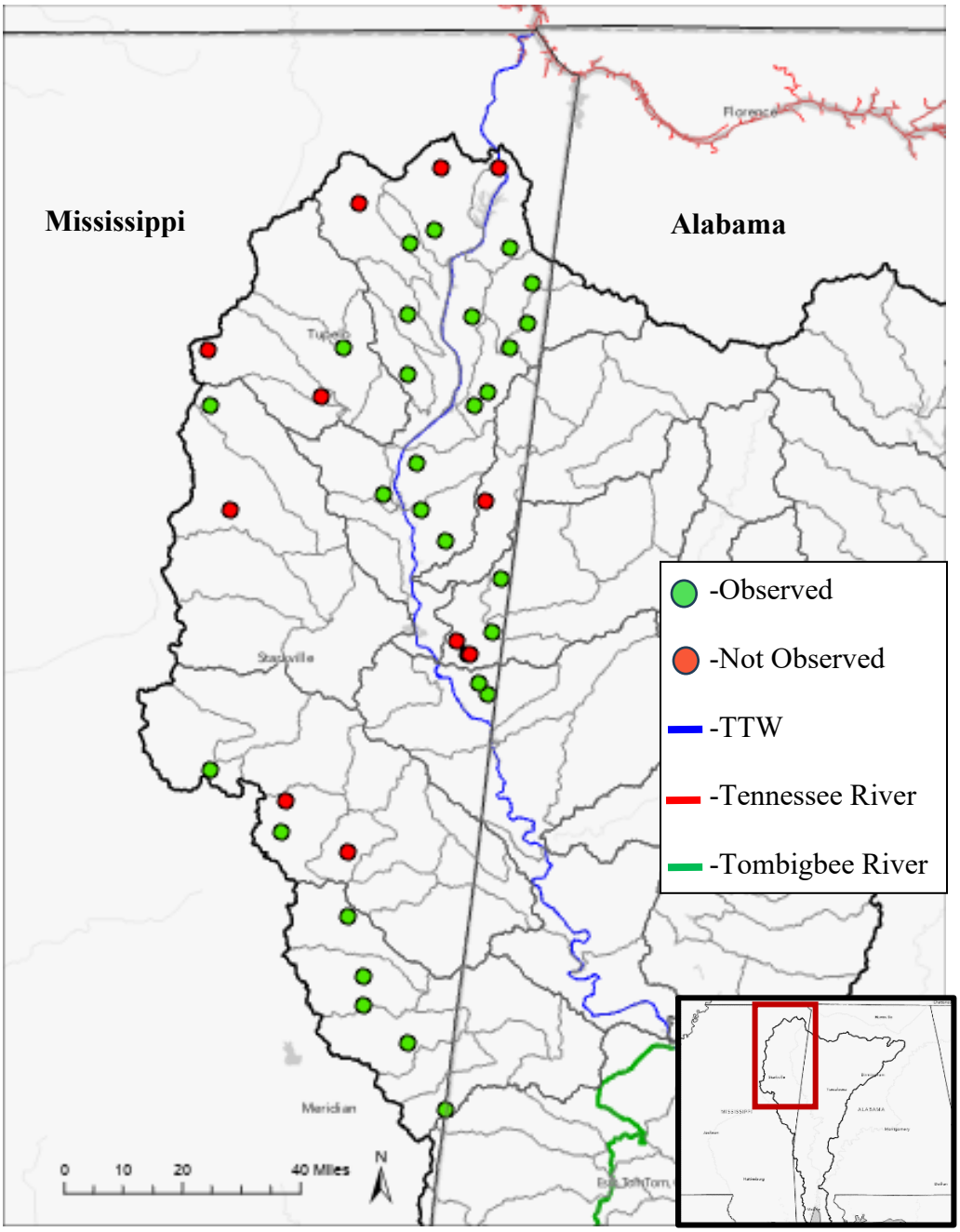

Figure 1.1 Map of sites surveyed in Northeast Mississippi, with colored dots representing if *E. lachneri* were observed at each site

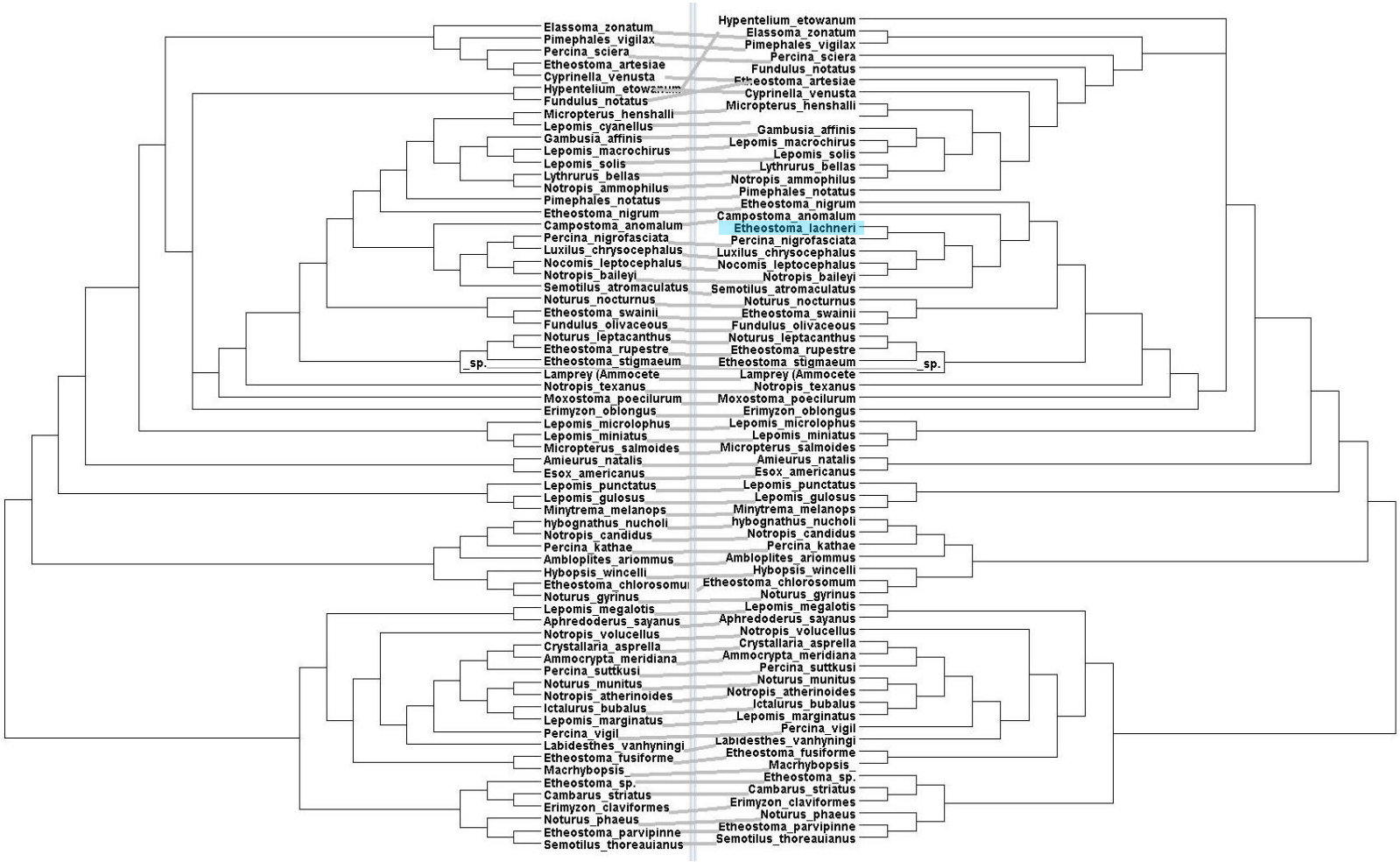

Figure 1.2 Tanglegram depicting how *E. lachneri* (highlighted in blue) present (on right) vs absent (on left) influences fish community relationships using Neighbor Joining Analyses.

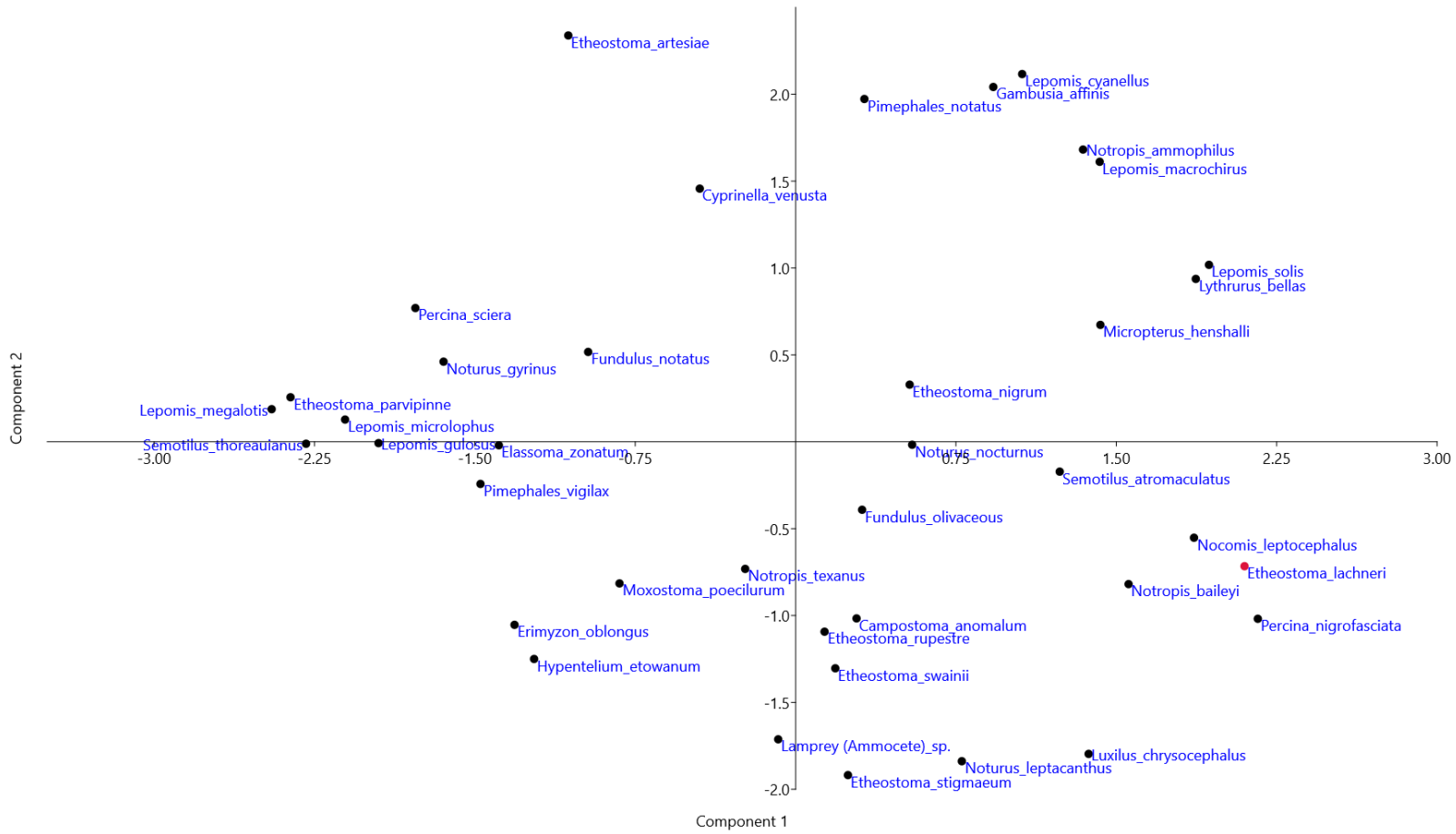

Figure 1.3 Principal Component Analysis (PCA) based on species occurrence data at 40 sampling sites using PAST™.

CHAPTER II

EVALUATING POPULATION GENOMICS OF THE TOMBIGBEE DARTER (*ETHEOSTOMA LACHNERI*)

2.1 Background

Although the southeast United States is inhabited by biodiverse aquatic communities and with many endemic species, most of this land is not protected and is leading to a steady decline in native ichthyofauna (Jenkins et al., 2015). Anthropogenic disturbances are the primary pressures leading to the decrease of aquatic diversity of fish assemblages (Johnston & Maceina, 2009). Previous surveys have shown less flora and fauna in this area due to agriculture runoff and changing of stream substrate (Barone & Hill, 2007).

The Mobile River Basin is incredibly biodiverse and home to many species of darter, including *E. lachneri* (Ross & Slack, 2000). *Etheostoma lachneri* was first put in the subgenus *Ulocentra* in 1999 (Porter, 1999). This subgenus is also known as snubnose darters and was created in 1878 (Jordan, 1878). Species within this subgenus are known for their allopatric speciation, which occurs when their home ranges do not overlap (Porter et al., 2002). The distribution of *E. lachneri* is one of the most widespread among the snubnose darters in the Gulf Coast drainage at approximately 15,000 km² (Suttkus et al., 1994). Its range spans across the Tombigbee River Watershed in the Mobile River Basin, which drains portions of Alabama and Mississippi (Mettee et al., 1989). Although *E. lachneri* has one of the largest geographical ranges

among snubnose darters in the Gulf Coast drainage, little information is available regarding population genetic structure across the landscape.

There is scant scientific literature on the habitat and life stages of *E. lachneri*, yet genetic research seems to be increasing for *E. lachneri* and other snubnose species (Kim et al., 2023; Kozal et al., 2017; Near et al., 2011; Sterling & Melvin L. Warren, 2020). Snubnose darters are a unique group of darters that are closely related and have a distinct downturned nose feature (Porter, 1999). Most snubnose species are listed as unknown, imperiled, or critically imperiled (Hammerson & Clausen, 2011). If the genetic diversity of *E. lachneri* populations is low, there could be a bottleneck effect, which occurs when population numbers are reduced and causes low genetic variation (Hänfling & Brandl, 1998). This could cause reduced adaptability and fitness levels of darter populations. While *E. lachneri* can only be found in Alabama and Mississippi, there are 26 species and 6 undescribed snubnose darter species found throughout the southeastern United States (Near et al., 2011). Historical isolation of populations among independent watersheds has caused distribution limitations in not only darters, but several other fish species (Bermingham & Avise, 1986). Although this group of darters shares a similar history, some darters in the subgenus *Ulocentra* show geological isolation of species (Powers & Warren, 2009). Additionally, other darters like *Etheostoma rupestre*, which occupies a similar habitat to *E. lachneri*, exhibit allopatric speciation, which is speciation due to geological isolation (Hilburn et al., 2023). *Etheostoma rupestre* is found in the same locality as *E. lachneri* and other snubnose species. Due to this, *E. lachneri* could be exhibiting allopatric speciation.

Based on the latest 2023 survey (see Chapter I), *E. lachneri* populations seem to be secure in Mississippi. However, genetic analyses could uncover underlying vulnerabilities that are not reflected in population counts alone. Another approach to assessing the health of a fish

population is through DNA sequencing, which can help define their genetic diversity (Martinez et al., 2018). With recent speciation occurring in the subgenus *Ulocentra*, *E. lachneri* DNA needs to be sequenced to see if populations are genetically stable. If there is no or low genetic diversity in *E. lachneri*, it could mean that the darters are more susceptible to environmental stressors and could potentially lead to extinction over time (Vrijenhoek, 1998).

This chapter uses next-generation sequencing to test the hypothesis that populations of *E. lachneri* are monophyletic. Alternatively, different populations may have speciated due to allopatry. Results of this study will enhance understanding of a darter species of conservation concern allowing for better management and conservation of endemic freshwater biodiversity.

2.2 Methods

2.2.1 DNA Extractions

Etheostoma lachneri specimens were collected in accordance with Mississippi permit #0531231 and stored in 95% ethanol (EtOH). Two Alabama Darter (*Etheostoma ramseyi)* specimens were collected at Mud Creek, AL (33.256572, 87.078623), in accordance with Alabama permit #11035, and used as an outgroup for genetic analyses. Sites that our survey did not cover (see Chapter I, Figure 1.2) were sampled by Ken Sterling (Okanogan-Wenatchee National Forest, Naches Ranger District, US Forest Service), who sent our lab tissues for analysis. *Etheostoma lachneri* caudal and right pectoral fins were used for DNA extraction using the DNeasy™ Blood and Tissue Kit (QIAGEN, Hilden, Germany). Extracted DNA was tested using a 1.5% agarose gel electrophoresis, and its concentration was evaluated using the NanoDrop 2000 Spectrophotometer™ to ensure the DNA concentration was high quality.

2.2.2 Sequencing Methods

The mitochondrial gene cytochrome *b* (cyt-*b*) was amplified using polymerase chain reaction (PCR) on a GeneAmp™ PCR System 9700 by Applied Biosystems and cleaned up using ExoSAP-IT™. To sequence the mitochondrial cyt-*b* gene, the amplification assay *Cytb_Percina,* modified by Dr. Brian Kreiser*,* contained 0.5 µL of the forward (cytbF_Percina - ATGGCTTGAAAAACCACCGTTG) and reverse (cytbR_Percina - CTCCGGCGTCCGGTTTACAAGAC) sequences, along with 23 µL of UltraPure™ DNase/RNase-free distilled water (Invitrogen™) and cytiva PuReTaq Ready-To-Go™ PCR Beads (Danaher™, Washington, D.C., USA), were used to isolate the desired amplicon for 72 *E. lachneri* (Song et al., 1998). A standard thermocycling profile was used, containing an initial hold of 94°C for 5 minutes, 35 cycles of denaturation at 90°C for 1 minute, annealing at 42°C for 1 minute, and elongation at 72°C for 2 minutes, and a final hold at 72°C for 7 minutes (Song et al., 1998). Each 25 µL PCR product was stored at 4°C until the cleaning step.

The PCR products were checked with gel electrophoresis to confirm amplification of the desired DNA fragment. After a successful gel check, 2.5 µL of ExoSAP-IT™ was added into the 22 µL PCR product to clean the DNA replication. A standard thermocycling profile of 37°C for 30 minutes and 80°C for 15 minutes was performed. After completion, the sample was held at 4°C until the sample was ready to be used or stored and checked again using gel electrophoresis. The DNA aliquots (N = 72) were shipped to Psomagen for cyt-*b* amplicon sequencing (Table B.1). *Etheostoma lachneri* cyt-*b* sequences from Psomagen were aligned to *E. lachneri* (GenBank ID: AF288436) from National Center for Biotechnology Information (NCBI) using MAFFT v.7 online server and uploaded onto NCBI (Katoh et al., 2002).

Etheostoma lachneri cyt-*b* sequences were mapped alongside HUC 8 watersheds, revealing that not all cyt-*b* haplogroups were topographically contiguous. To assess geographic patterns of genetic differentiation, nuclear DNA was sequenced. The DNA aliquots (N = 94), containing *E. lachneri* (N = 92) and *E. ramseyi* (N = 2), were shipped to Diversity Arrays Technology Pty Ltd (Canberra, Australia) for genotyping using DArTseq™ technology, which is a form of Double Digest Restriction-site Associated DNA Sequencing (ddRADseq; Table B.2). Individuals were selected from the following HUC 8 streams: Upper Tombigbee (N=7), Town (N=8), Tibbee (N=6), Buttahatchee (N=6), Luxapallila (N=7), Sipsey (N=5), Noxubee (N=8), Middle Tombigbee-Lubbub (N=8), Lower Black Warrior (N=8), Sucarnoochee (N=8), Middle Tombigbee-Chicasaw (N=8), Mobile-Tensaw (N=6), and Lower Tombigbee (N=7). To ensure SNP data would be high quality, additional individuals were selected based on higher nanodrop concentrations. Using Genius Prime™, SNPs from DArTseq™ were aligned and assembled.

2.2.3 Phylogenetic Analysis

A mitochondrial DNA sequence from Psomagen was uploaded to the CIPRES Science Gateway where IQTREE was chosen under the maximum-likelihood optimality criterion for phylogenetic analysis. The resulting tree file was visualized using the program FigTree v.1.4.4 (Rambaut, 2016). *E. colorosum* (GenBank ID: Hq128112, HQ128113), *E. bellator* (GenBank ID: HQ128088), *E. chermocki* (GenBank ID: HQ128103), *E. brevirostrum* (GenBank ID: HQ128096, HQ128097), *E. tallapoosae* (GenBank ID: HQ128241, HQ128242), *E. ramseyi* (GenBank ID: HQ128211, HQ128212), *E. raneyi* (GenBank ID: HQ128213, HQ128214), *E. cervus* (GenBank ID: HQ128101, HQ128102), *E. pyrrhogaster* (GenBank ID: HQ128206, HQ128207), *E. zonistium* (GenBank ID: HQ128255, HQ128256), *E. etnieri* (GenBank ID: HQ128128), *E. baileyi* (GenBank ID: HQ128081, HQ128082), *E. coosae* (GenBank ID:

HQ128114, HQ128115), *E. scotti* (GenBank ID: HQ128218), *E. duryi* (GenBank ID: HQ128123, HQ128124), and *E. flavum* (GenBank ID: HQ128132, HQ128133) were used as outgroups from the National Center for Biotechnology Information (NCBI) online server. A pair-wise distance analysis was conducted in the program MEGA v.12.0.11 using cyt-*b* sequences to analyze haplogroup differences (Kumar et al., 2024).

2.2.4 Genetic Analysis

Binary SNP data from DArTseq™ were uploaded onto R® (R Core Team, 2024) in the RStudio® interface (v.2024.04.2) along with a metadata file containing coordinates, population name, and the stream that the individual was collected. Packages *dartR* (Gruber et al., 2018), *adegenet* (Jombart, 2008), *dartRverse,* and *LEA* (Frichot & François, 2015), were used to evaluate population genomics. The SNP data was formatted to a genlight object and filtered at a 90% call rate and reproducibility of loci. Additionally, monomorphic alleles and secondary loci were filtered from the SNP data. A Hardy-Weinberg test was conducted on the filtered data using the function *gl.hwe.pop* in the *dartR* package, to look for potential ohnologs (Hosking et al., 2004). The resulting filtered SNP sequence was generated into a FASTA file and exported for phylogenetic analysis. The DNA alignment was uploaded to the CIPRES Science Gateway, where IQTREE was chosen under the maximum-likelihood optimality criterion for phylogenetic analysis. The resulting tree file was visualized using the program FigTree v.1.4.4 (Rambaut, 2016).

A *LEA* ancestry analysis conveys which individuals fall in which population along with add-mixed individuals and their percentages (Frichot & François, 2015; Zbinden et al., 2023). To examine ancestral lineages, an ancestral analysis was run in RStudio® using the *LEA* package. The file was then exported as a comma-separated values (.csv) and visualized in Microsoft®

Excel®. The number of classifications represented by K, was determined by a cross-entropy analysis which included *E. ramseyi* as the outgroup. Cross-entropy was run with a range (K=3-11) along with fifteen replicates for each value of K. Additionally, a maximum likelihood phylogenetic tree containing individuals with < 30% admixture from the *LEA* ancestry analysis was created to see if phylogenetic grouping would be altered (Figure B.1). To examine relationships between populations across geographical regions, a principal coordinate analysis (PCoA) was performed using *dartR* to visualize genetic clustering using the command *gl.pcoa* then plotted using *gl.pcoa.plot* (Wilkinson & Edds, 2001). Fixation Index (F_{ST}) measured genetic variation between populations on a scale of 0 to 1, where 0 represented sharing the same alleles and 1 meant no shared alleles. In the ichthyology community, a F_{ST} value over 0.20 represents species-level differences between populations (Wright, 1984). The F_{ST} was calculated using populations from the *LEA* ancestry analysis using the command *gl.fst.pop* in the *dartR* package. One thousand bootstrap replicates were used to perform F_{ST} across loci, in addition to a 95% confidence interval.

Bayesian Phylogenetics & Phylogeography (BPP) and Genealogical Diversity Index were used to align population and multi-species DNA to show a dependable phylogenetic relationship (Kim et al., 2022). RStudio®, Python®, Windows Command Prompt®, Tracer, Minimalist BPP, Notepad®, and FigTree were used to calculate and align Bayesian Phylogenetics & Phylogeography (BPP) and Genealogical Diversity Index (gdi). One hundred loci were randomly selected for eight individuals in each of the three genetic groups: Upper Tombigbee, Lower Tombigbee, and Lower Black Warrior. For BPP and gdi analyses, SNPs were filtered with a call rate of 1.0. For all analyses in BPP, the control file was run with burnin=50,000, sample frequency=2, nsample=150,000, threads=2, and scaling=1. *Etheostoma ramseyi* was used as the

outgroup to calculate the species tree using the BPP analysis A01. The A10 analysis was used to perform a species delimitation. To calculate estimates of tau and theta, analysis A00 was run for parameter estimation.

2.2.5 Population Genomic Statistics

Inbreeding coefficient (FIS), expected heterozygosity (H_e), and observed heterozygosity (H_o) determine the presence of gene flow and inbreeding within populations (Weir & Cockerham, 1984). The FIS measures genetic variation within the population. The scale for FIS ranges from 0 to 1, with 0 indicating little to no inbreeding and 1 indicating high inbreeding within the population. Using SNP data, FIS, H_e, and H_o were evaluated with *gl.report.heterozygosity* in the *dartR* package in RStudio. Effective population size (N_e) estimates the number of individuals in that population that would hypothetically have the same genetic drift as the actual population (Wright, 1938). Effective population size (N_e) was estimated for *E. lachneri* using the software *NeEstimator* v2 using the *gl.LDNE* command in *dartR* with a lowest allele frequency of 0.010 used along with random mating (Do et al., 2014).

For isolation by distance (IBD), the Mantel test was used to examine correlations between population variation and geographical distance (Camak & Piller, 2018). For IBD, a csv file that contained F_{ST} values of genetic distance and locality was imported into RStudio® and ran with a river network using the package *riverdist*. Genetic distance and geographic distance were then turned into a matrix using the *as.matrix* command. The distances from each matrix were calculated using the *dist* command. A *mantel.randtest* was then performed on the matrices distances to see the correlation between the two distance matrices. The matrices distances were then turned into vectors and plotted along with geographical distance represented by the x-axis and genetic distance represented as the y-axis using *ggplot*.

2.3 Results

2.3.1 Phylogenetic sequencing

A maximum likelihood phylogenetic tree for the cyt-*b* gene shows four clades with high bootstrap support (Figure 2.1). Three clades exhibit geographic structure (i.e., list the three clades) while the green clade represents individuals that have a widespread distribution and lack structure (Figure 2.2). Pair-wise distances between genetic haplogroups show high genetic similarities between groups (Table 2.1)

For SNP data, our 92 *E. lachneri* samples had 71,491 SNPs with 39.98% missing data. After filtering, SNP sequences had 11,171 SNPs with 3.06% missing data. For nuclear sequencing, a maximum likelihood phylogenetic tree shows three evolutionarily significant units (ESUs) with bootstrap values over 96 for each (Figure 2.3). Nuclear DNA showed three distinct ESUs populations: Black Warrior-Tombigbee (Upper Tombigbee), Mobile Bay-Tombigbee (Lower Tombigbee), and Lower Black Warrior. Black Warrior-Tombigbee and Mobile Bay-Tombigbee refer to the Hydrologic Unit Code (HUC) 6 for the area, while Lower Black Warrior refers to the HUC 8. Populations were mapped using ArcGIS Pro™ and showed that populations were geographically contiguous (Figure 2.4).

2.3.2 Population Genomic Analyses

The *LEA* ancestral analyses with *E. ramseyi* included as the outgroup showed that K=4 was the best solution and agreed with the maximum likelihood phylogenetic tree (Figure 2.3). The *LEA* ancestry analysis shows three ESUs in addition to *E. ramseyi* as the outgroup (Figure 2.3). The resulting three ESU's of *E. lachneri* from the *LEA* ancestry analysis [Upper Tombigbee (mean=0.977, sd= 0.038), Lower Black Warrior (mean=0.989, sd=0.019), and Lower Tombigbee (mean=0.934, sd=0.061)] had some admixture depicted by different coloration within their main

color. The maximum likelihood phylogenetic tree shows that individuals from Sipsey are genetically distinct from the Upper Tombigbee population (Figure B.1). However, upon further investigation using a *LEA* ancestral plot, the Sipsey population has >30% admixed individuals, which likely explains the position of this population on the maximum likelihood phylogeny (Figure 2.3).

The PCoA results revealed three clusters including: Lower Tombigbee, Upper Tombigbee, and Lower Black Warrior (Figure 2.6). These results support the *LEA* ancestral plot. The PCoA axes represented variation between individuals: PCA axis 1 at 27.2% and PCA axis 2 at 10.9% (Figure 2.6).

Upper Tombigbee, when compared to Lower Tombigbee, resulted in a lower F_{ST} value (0.43), than when compared to Lower Black Warrior (0.6; Table 2.1). Lower Tombigbee compared to Lower Black Warrior had a F_{ST} slightly higher than Upper Tombigbee compared to Lower Tombigbee (0.46). The fixation index value among Upper Tombigbee, Lower Tombigbee, and Lower Black Warrior had values suggesting there are species level differences (Table 2.2).

All three ESU's were delimited using gdi, which were determined by the phylogenetic analysis. Upper and Lower Tombigbee populations were in the ambiguous range, both with a gdi value of 0.4, meaning weak support for divergence (Figure 2.7; Jackson et al., 2017). The Lower Black Warrior population was in the upper bound with a gdi value of around 0.85, meaning there is high support for speciation (Figure 2.7). While not all ESUs were in the high support range, supporting evidence from our phylogenetic tree, PCoA, and *LEA* indicate speciation. The minimum threshold of gdi for divergence is 0.20, which all ESUs exceeded (Jackson et al., 2017; Leaché et al., 2019). Our results showed that the Lower Black Warrior, Lower Tombigbee, and

Upper Tombigbee population had H_e values higher than H_o values and FIS values over 0.5 (Table 2.3; Figure 2.8). Effective population size (N_e) results showed that populations in the Upper Tombigbee, Lower Tombigbee and Lower Black Warrior all had $N_e > 10$ (Table 2.4). The analysis of IBD suggested that while there was a significant positive correlation between genetic and geographic distance, only a moderate amount of genetic variation was explained by river distance ($P < 0.05$; $R^2 = 0.52$; Figure 2.9).

2.4 Discussion

2.4.1 Phylogenomic Structure

While cyt-*b* indicates low genetic differences of haplogroups, SNP results show three genetically distinct populations of *E. lachneri* in the Upper Tombigbee, Lower Tombigbee, and Lower Black Warrior, potentially indicating a speciation event. The population referred to in this paper as Lower Tombigbee is in the locality where *E. lachneri* was described (Suttkus et al., 1994). The results violate the null hypothesis that populations of *E. lachneri* are panmictic since there are three genetically distinct populations. The ESUs plotted on a map along with HUC showed that the clades were topographically contiguous except for one individual, 10072, that was in the Lower Tombigbee ESUs population but was in the Upper Tombigbee watershed (Figure 2.4). Upon further investigation using the National Hydrography Dataset (NHD) Flowlines from the United States Geological Survey (USGS), it is believed this individual could have traveled due to a stream capture event (Jarrett et al., 2017; Masuda et al., 2023; Strange, 1998; U.S. Geological Survey, 2019).

2.4.2 Support for Species Complex

All three ESUs populations of *E. lachneri* show distinct ancestral lineages with little to no admixture among populations, meaning there could be a physical or ecological barrier separating populations from interbreeding. Our PCoA and F_{ST} results indicate support for three ESUs populations of *E. lachneri* with little genetic mixing among the 3 groups (Table 2.2; Figure 2.6).

The results of this study convey that there are geographical or physical barriers present that could influence allopatric speciation. Other darter species such as *E. rupestre* have been influenced by barriers that have resulted in allopatric speciation (Hilburn et al., 2023). While gdi results show that Upper and Lower Tombigbee have weak support for divergence, other analyses such as the phylogenetic tree, *LEA* ancestry, and PCoA support possible speciation. When calculating expected heterozygosity and observed heterozygosity, if the H_o is lower than the H_e there is an inconsistency, which could imply inbreeding (Hänfling & Brandl, 1998). High inbreeding within a population can induce decreases in immune response to diseases, growth rate, and fecundity, which can end up causing extirpation in fish populations (Reed & Frankham, 2003). All three populations had H_o lower than H_e, revealing low genetic diversity within each ESU. (Table 2.3; Figure 2.8). Relative to comparable reports for members of *Etheostoma* species, the effective population sizes of *E. lachneri* populations are exceptionally low (Fluker et al., 2010; Table 2.4). Lower Black Warrior had the lowest N_e value, which may be due to only having eight individuals that were geographically near each other. The IBD of *E. lachneri* shows a statistically significant positive correlation between F_{ST} and geographical distance (River km; Figure 2.9).

2.4.3 Conservation and Management Implications

Given genetic support from *E. lachneri* for two new species, morphology work needs to be conducted to determine physical differences within the three populations. These results are supported by *LEA* ancestry, PCoA, F_{ST}, gdi, and IBD. Additionally, if these populations are distinct species, these new species need their population status to be reevaluated. Each of these populations have a high FIS, representing a greater risk of extirpation for these individuals. In addition, N_e estimator shows that few individuals are breeding in each population meaning that management needs to be implemented. While these population genetics are positively correlated with distance, populations within the Lower and Upper Tombigbee are physically separated by the Tennessee Tombigbee Waterway. Previous studies have shown that other snubnose darters such as *E. raneyi* exhibit similar conditions in Mississippi and need management and potential breeding programs (Sterling et al., 2020).

2.4.4 Conclusion

Results presented here provide strong evidence that *E. lachneri* is comprised of three monophyletic and geographically contiguous ESUs. According to the Phylogenetic Species Concept, species descriptions must provide evidence of monophyly and diagnosability (Cracraft, 1983; De Queiroz & Donoghue, 1990). While the Upper Black Warrior and Upper Tombigbee populations are monophyletic, morphological diagnosis will require additional data not presented here. Recent species descriptions have interpreted collection locality (watershed) satisfies the diagnosability criterion, but in this study, this approach can be considered premature (Egge & Simons, 2006; Young et al., 2013). This study provides a phylogenetic framework that should be leveraged in the design of morphological comparisons in the future. In lieu of morphological

data, this study also provides conservation agencies with valuable information that can be used to conserve genetic diversity within each ESU.

2.5 Tables

Table 1.6 Pair-wise distances between genetic haplogroups of *Etheostoma lachneri* mitochondrial cytochrome *b* sequences.

Haplogroup	**Everywhere**	**Sucarnoochee**	**Mobile-Tensaw**	**Lower Black Warrior**
Sucarnoochee	0.033			
Mobile-Tensaw	0.016	0.028		
Lower Black Warrior	0.021	0.031	0.0156	

Table 1.7 Fixation Index (F_{ST}) for three populations of *E. lachneri* inferred from *LEA* ancestry and phylogenetic analyses using single-nucleotide polymorphism (SNP) sequencing. A threshold over 0.2 implies species level differences (Wright, 1984).

Population	**Upper Tombigbee**	**Lower Tombigbee**	**Lower Black Warrior**
Upper Tombigbee	NA	NA	NA
Lower Tombigbee	0.43	NA	NA
Lower Black Warrior	0.60	0.46	NA

Table 2.1 Observed Heterozygosity, Expected Heterozygosity, and Fixation Index values for three populations of *E. lachneri* inferred from *LEA* ancestry and phylogenetic analyses using single-nucleotide polymorphism sequencing.

Population	**nInd**	**He**	**HeSD**	**H_O**	**HoSD**	**FIS**
Lower Black Warrior	7.55	0.06	0.15	0.03	0.10	0.50
Lower Tombigbee	27.87	0.14	0.17	0.06	0.09	0.60
Upper Tombigbee	53.86	0.09	0.15	0.04	0.08	0.57

Abbreviations: Number of Individuals (nInd), Expected Heterozygosity (He), Expected Heterozygosity Standard Deviation (HeSD), Observed Heterozygosity (HO), Observed Heterozygosity Standard Deviation (HOSD), Inbreeding coefficient (FIS).

Table 1.8 Effective population size (Ne) for *E. lachneri* among three evolutionarily significant units populations inferred from *LEA* ancestry and phylogenetic analyses using single-nucleotide polymorphism sequencing.

Population	**Sample Size (n)**	**Estimated Ne**	**JackKnife 95% CI**
Lower Black Warrior	8	2.6	1.4-11.8
Lower Tombigbee	29	4.5	3.3-9.4
Upper Tombigbee	55	9.4	6.7-12.3

2.6 Figures

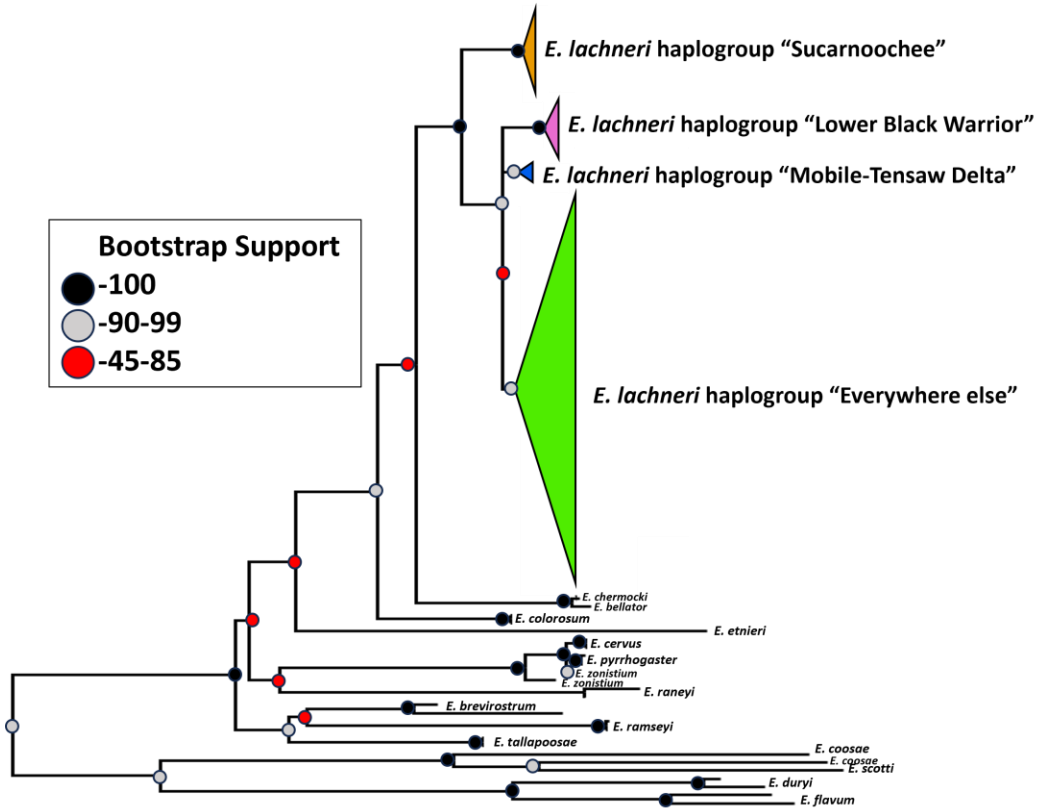

Figure 1.4 Mitochondrial cytochrome *b* gene maximum likelihood phylogenetic tree for 72 *E. lachneri* with nodes and bootstrap support values present (substitution model: TIM3+F+I+G4). Sucarnoochee (Orange), Lower Black Warrior (Pink), Mobile-Tensaw Delta (Blue), Everywhere else (Green) shown as collapsed clades. Snubnose darters (n=28) used as outgroups were not collapsed.

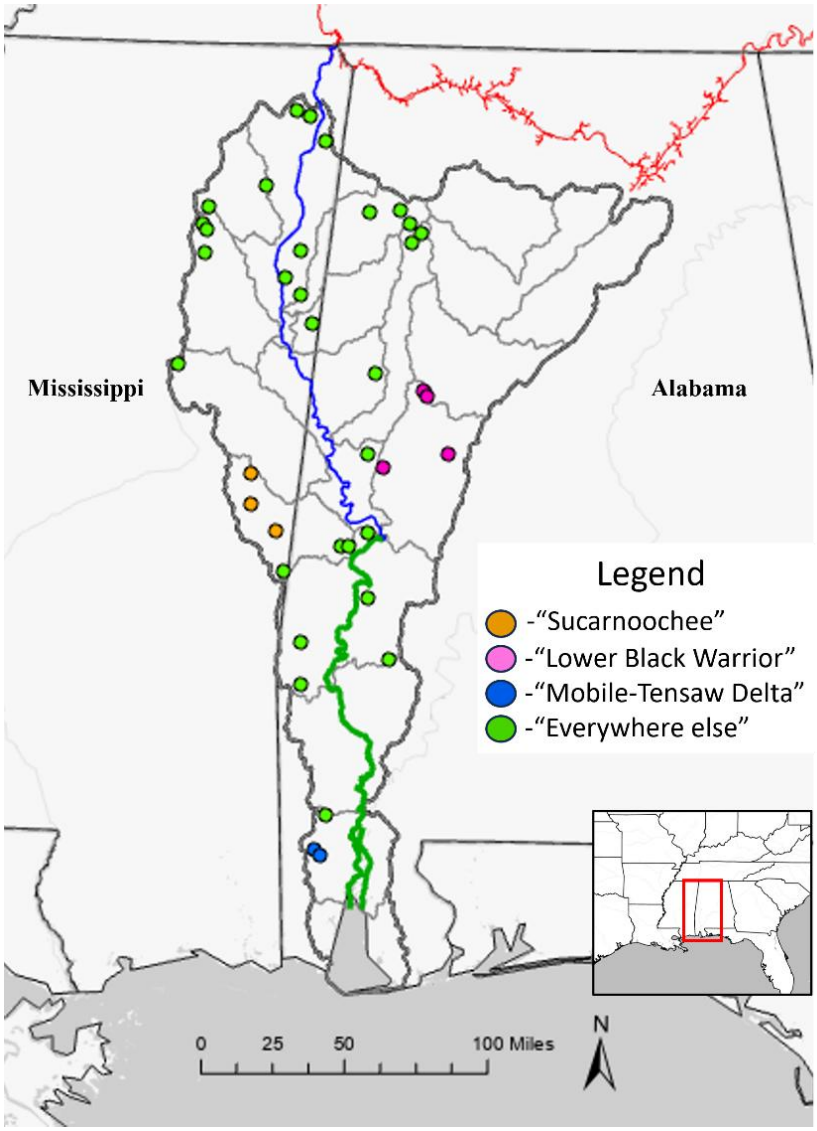

Figure 1.5 Map of mitochondrial cytochrome *b* gene maximum likelihood phylogenetic tree results for 76 *E. lachneri* with dots representing clades.

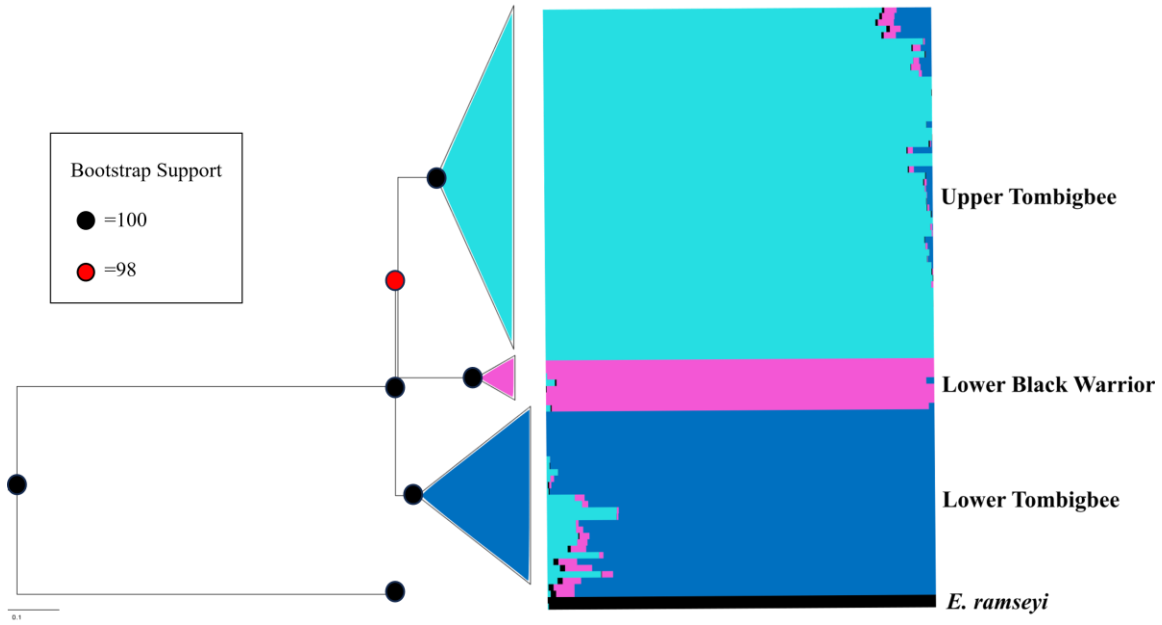

Figure 1.6 Maximum-likelihood phylogenetic tree (substitution model: TVM+F+G4) and *LEA* ancestry plot (K=4) for *E. lachneri* using 92 single-nucleotide polymorphism sequencing with 2 *E. ramseyi* (Black) used as an outgroup and bootstrap support values present. Upper Tombigbee (Turquoise), Lower Black Warrior (Pink), and Lower Tombigbee (Blue) shown as collapsed clades are names that represent the three evolutionarily significant unit populations.

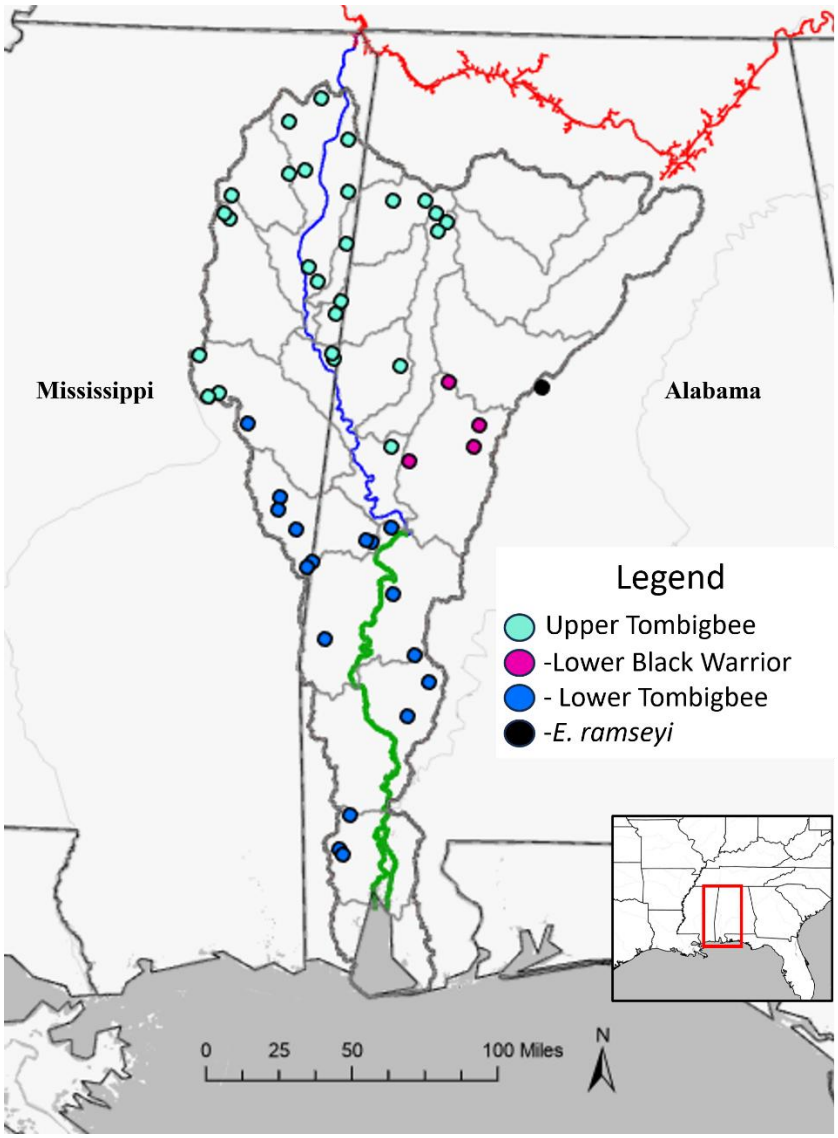

Figure 1.7 Map of *E. lachneri* single nucleotide polymorphism maximum likelihood phylogenetic tree with dot color representing the three evolutionarily significant units, populations, and outgroup.

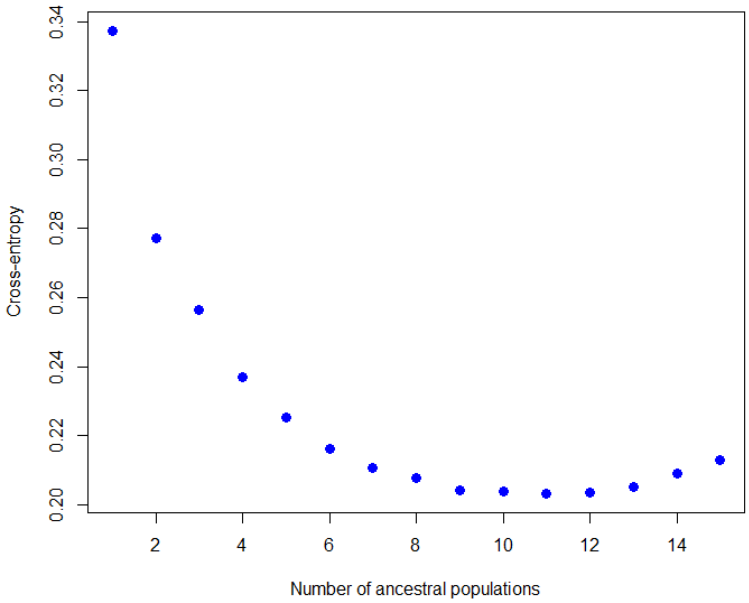

Figure 1.8 Cross-entropy plot representing the best fit of ancestral populations of filtered single nucleotide polymorphism (SNP) of *E. lachneri.*

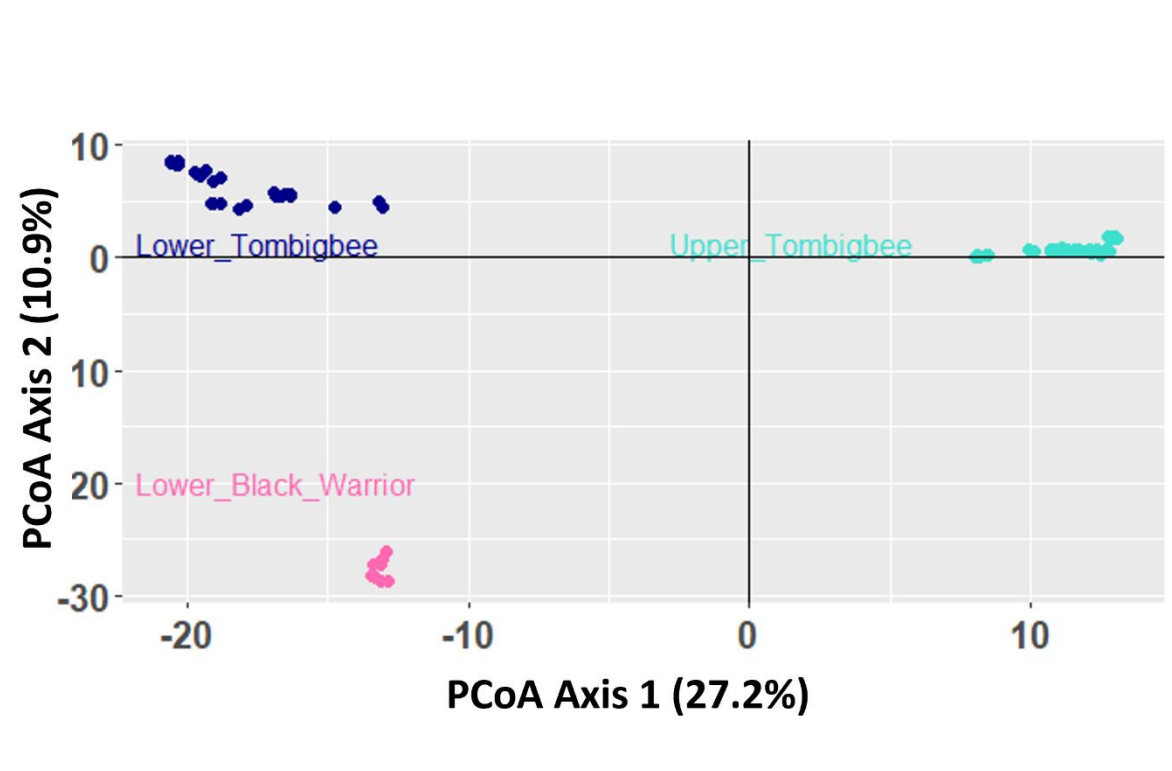

Figure 1.9 Principal coordinate analysis (PCoA) for the three evolutionarily significant units populations of *E. lachneri* inferred from *LEA* ancestry and phylogenetic analyses using single-nucleotide polymorphism sequencing.

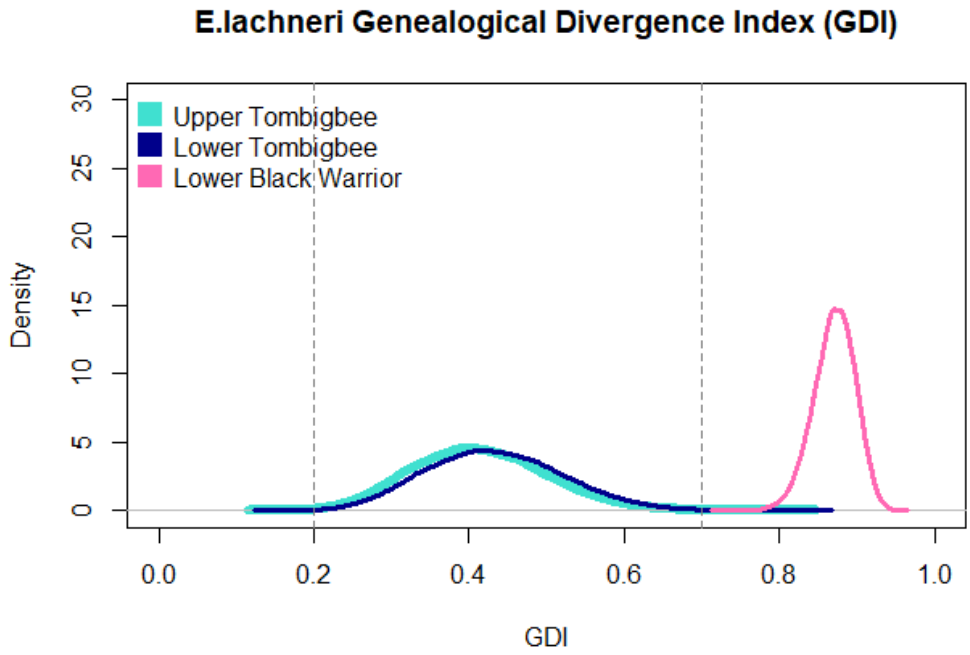

Figure 1.10 Genealogical diversity index (gdi) for the three evolutionarily significant units populations of *E. lachneri* inferred from *LEA* ancestry and phylogenetic analyses using single-nucleotide polymorphism sequencing.

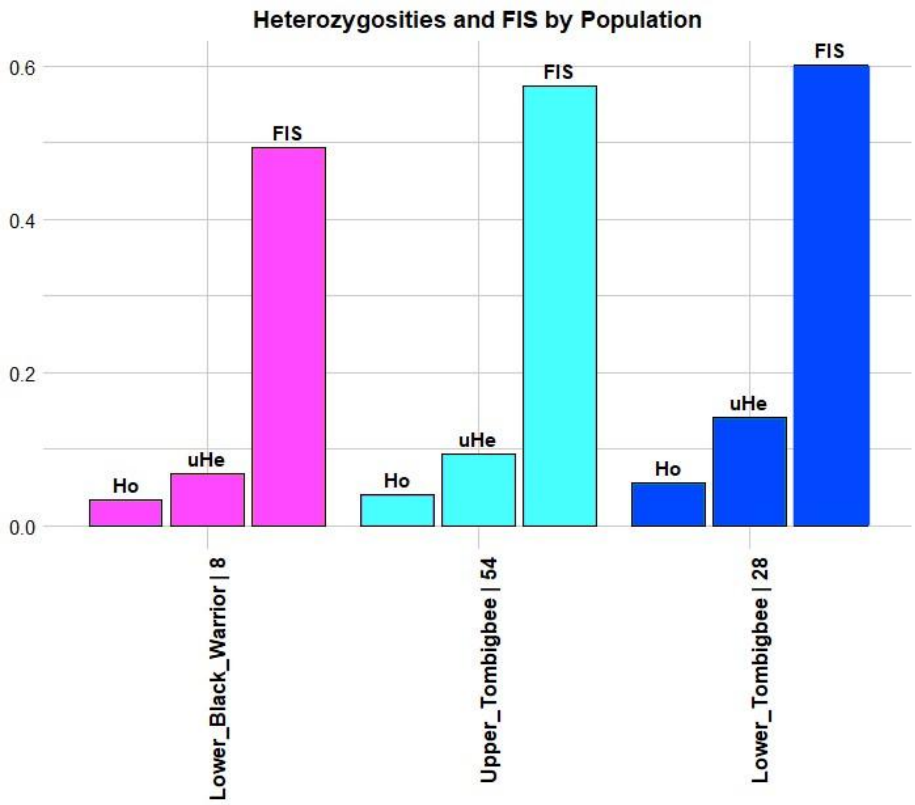

Figure 1.11 Heterozygosity and Inbreeding coefficient (FIS) bar plot for three evolutionarily significant units populations of *E. lachneri* from single-nucleotide polymorphism sequencing.

Abbreviations: Observed Heterozygosity (H_O) and Expected Heterozygosity (uH_e).

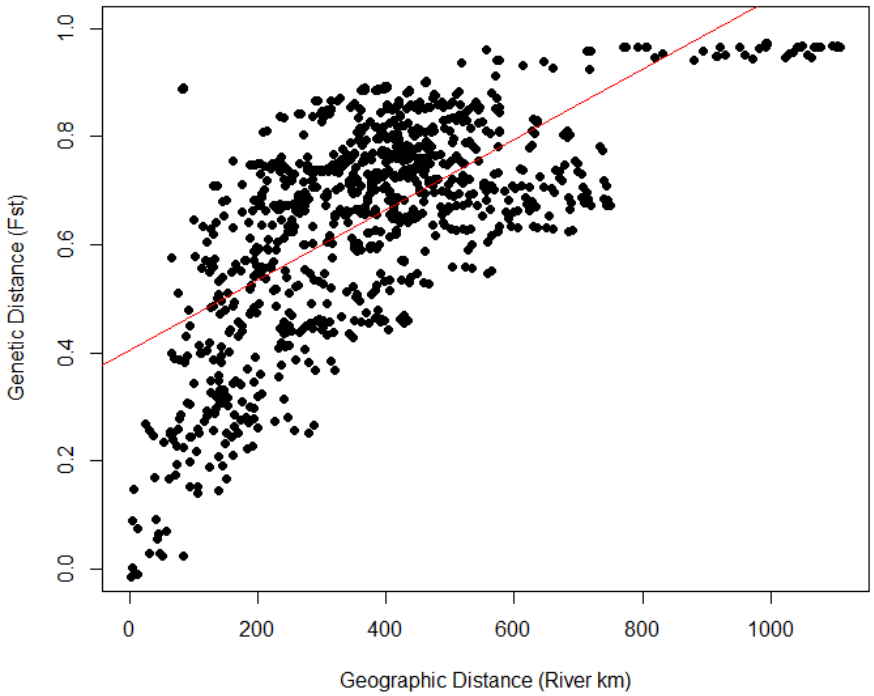

Figure 1.12 Graph showing isolation by distance (IBD) comparing genetic and geographical distance of *E. lachneri* single-nucleotide polymorphism (SNP) sequencing (P = 0.001; $R^2 = 0.52$).

CHAPTER III

DETECTION OF TOMBIGBEE DARTER (*ETHEOSTOMA LACHNER*I) USING ENVIRONMENTAL DNA (EDNA)

3.1 Background

3.1.1 Environmental DNA

Environmental DNA (eDNA) is an increasingly popular method of species detection by collecting DNA from water, air, and soil. This newer method is cost-effective and non-invasive to the environment, compared to traditional sampling methods such as electrofishing, dipnetting, seining, and trawling (Dejean et al., 2012; Rees et al., 2014). This project includes the collection of eDNA in addition to a stream survey to detect *E. lachneri* that may not have been collected during traditional surveys. These traditional surveys used a Smith-Root® LR-24 backpack electrofisher (Smith-Root, Vancouver,Washington) and seine nets for the methods and were conducted in 2023 across tributaries of the Tennessee-Tombigbee Waterway in Mississippi (see Chapter I). Since populations of *E. lachneri* are known to have low densities at some sites (see Chapter I), eDNA may be an efficient tool for detecting the presence of these low abundance taxa (Bohmann et al., 2014; Carvalho et al., 2021; Valentini et al., 2016; Wang et al., 2016). Currently, the execution of eDNA protocols can be difficult due to contamination, data interpretation, and the unknown abundance of species (Beng & Corlett, 2020; Jerde, 2021). Quantitative polymerase chain reaction (qPCR) was performed to quantify the relative abundance of *E. lachneri* from eDNA samples.

Because some species of *Ulocentra* are threatened and endangered, the availability of subspecies specific eDNA primers would be beneficial for the multitude (IUCN, 2025). Species of *Ulocentra* spp. do not co-occur. This renders *Ulocentra* specific primers effective for species detection without the risk of congener cross-amplification.

3.1.2 Tradeoffs

While eDNA methods offer a cost and time-effective alternative to traditional field sampling, there are limitations to quantifying the number of individuals in a survey area (Yates et al., 2025). When imperiled or potentially imperiled taxa are present, there is a need to estimate abundance in order to identify and rank the most at-risk populations (Adams et al., 2019; Cowart et al., 2018). Quantifying fish abundance with eDNA can be a powerful tool, yet studies have shown that DNA presence can be affected by abiotic and biotic factors (Lacoursière-Roussel et al., 2016; Wang, 2017). While eDNA has limitations compared to field sampling, methods and research are optimizing it over time.

This chapter uses standardized eDNA sampling methods in 39 wadeable streams in Mississippi Tennessee Tombigbee Waterway tributaries to evaluate *E. lachneri* detection. The results of this study will provide eDNA primers and probe to detect *E. lachneri* in water and allow a comparison of the detection of *E. lachneri* using eDNA versus traditional surveying as described in Chapter 1.

3.2 Methods

3.2.1 Filter extractions

During the field surveys at thirty-nine sites described in Chapter I (Table 3.1), 1L of water was filtered using the Grover-Pro™ eDNA pump (Grover Scientific Pty Ltd). Given the variability of turbidity across sites, filters with different pore sizes were used to ensure proper filtration (Nolan et al., 2023). The range of Smith-Root™ self-preserving filter (Smith-Root, Vancouver, Washington) pore sizes employed varied from 0.45 µM to 5 µM. Water was filtered near bridges and culverts upon first access to sites in an effort to decrease contamination. One liter of distilled water was filtered on-site and used as a field negative control to test for contamination. The self-preserving filters were stored at room temperature following the manufacturer's suggestion.

We extracted eDNA using the DNeasy PowerWater Kit™ (QIAGEN), following manufacturer instructions. All lab benches and tools used for extractions were decontaminated using a 10% bleach solution. One half of each filter was extracted, and the other half was stored in 100% ethanol (EtOH) at -20°C as backup in case of re-extraction. Environmental DNA presence was confirmed using methods and universal vertebrate PCR primers (Alley et al., 2025). To determine if DNA was amplified in the PCR reaction, gel electrophoresis was used. Samples that did not amplify were dehydrated using a SpeedVac, rehydrated with 30 µL of UltraPure™ DNase/RNase-free distilled water (Invitrogen™), and rerun in a PCR reaction again.

3.2.2 qPCR Primer design

To aid in species-specific qPCR primer design, the complete mitogenome of one specimen of *E. lachneri* was sequenced and annotated. The specimen was collected from Mackeys Creek, MS (34.4718377, -88.260904) and was preserved at Mississippi State

University in 100% EtOH. Whole genomic DNA was extracted using the DNeasy Blood and Tissue Kit™ (QIAGEN, Hilden, Germany), and the quality was confirmed using gel electrophoresis with a 1.5% agarose gel that was stained with ethidium bromide. The DNA quantity was measured using a NanoDrop 2000 Spectrophotometer™ (Thermo Fisher Scientific, Waltham, MA, USA).

Genomic DNA was sequenced through Plasmindsaurus using Oxford Nanopore Technology (Oxford Nanopore, Oxford, UK). The mitogenome was assembled using Geneious Prime v.2021.1 with default parameters. Mitochondrial reads were isolated from nuclear reads by assembling to a reference (*Etheostoma tallapoosae*; GenBank ID: KY952221). MitoAnnotator v.3.65 was used for genome annotation.

The *E. lachneri* mitogenome (GenBank ID: PP915961) was aligned with mitogenomes of congeneric species using MAFFT v.7 (Katoh et al., 2002; Katoh & Standley, 2013). This alignment was viewed in Bioedit v.7.7.1 to identify areas of the mitogenome suitable for qPCR primer design (i.e., regions conserved within *Ulocentra* but variable in other co-occurring darters; Hall, 1999). An appropriate region was found to be present within the ND6 gene. Forward and reverse primers and an internal oligo were designed using Primer3Web v.4.1.0. An initial specificity validation was conducted using NCBI Primer-BLAST™ to ensure other co-occurring darters would not cross-amplify. Additional primer validation was completed by conducting conventional PCR using DNA from closely related and co-occurring darters (*E. lachneri, E. ramseyi*, *E. nigrum*, and *E. rupestre*) based on the traditional survey described in Chapter I. Thermocycling conditions were set for 95°C for 2 minutes, followed by 35 cycles of 94°C for 30 seconds, 56°C for 45 seconds, and 72°C for 1 minute, then held at 72°C for 10 minutes for one cycle. The PCR assay contained 0.5µL of the forward (*E. lachneri*_F_ND6 -

gctttgctggcctctattttagtac) and reverse (*E.lachneri*_R_ND6 - atcctagggctttgttctctagc) sequences, along with 21 µL of UltraPure™ DNase/RNase-free distilled water (Invitrogen™) and cytiva PuReTaq Ready-To-Go™ PCR. The PCR amplification was visualized using gel electrophoresis as described above.

3.2.3 qPCR

Quantitative polymerase chain reaction (qPCR) using the reporter BHQ1 was used to amplify and quantify *Ulocentra* DNA from eDNA. The PerfeCTaq qPCR ToughMix™ (Quantabio; Beverly, MA, USA) was used because of its ability to amplify target DNA in the presence of PCR inhibitors. Reactions were performed on an Agilent Technologies Stratagene Mx3005P™ using the program MxPro-Mx3005P™. Thermocycling conditions were set to 95°C for 10 minutes followed by 45 cycles of 95°C for 15 seconds and 62°C for 1 minute. The qPCR assay consisted of a total volume of 20µL including 1µL of diluted gBlock, 10µL of PerfeCTa qPCR ToughMix™, 900 nM of each primer, and 250 nM of internal probe. The rest of the solution was filled with UltraPure™ water.

A standard curve dilution series was created using a synthetic gBlock. The gBlock was ordered from IDT™ (Integrated DNA Technologies, Coralville, IA, USA) and designed using the forward and reverse primers to extract the target sequence from the sequence alignment. The volumes required to dilute the gBlock to the desired concentration were calculated using the Thermo Fisher Scientific DNA Copy Number and Dilution Calculator™. The gBlock (10 ng/µL) was diluted to 10^9 copies/µL. The standard curve consisted of six steps in a 10-fold serial dilution (10^6-10^1 copies/µL).

The limit of detection (LOD) and limit of quantification (LOQ) were calculated by performing qPCR on the dilution series described above. The LOD defines the least amount of

DNA that can be detected above a set threshold, while LOQ defines the least amount of DNA that can be quantified accurately above a set threshold. Filtered pipette tips were used to decrease the possibility of contamination. Filter set gain settings were adjusted accordingly for the fluorescent dye FAM (x4) and reference dye ROX (x1). Primer and probe concentrations were optimized using serial dilution, along with a temperature gradient for annealing temperature. Dilutions were run with three replicates in addition to three no-template controls (NTCs) per plate. A standard curve was optimized with an R^2 value of at least 0.98 and efficiency ranging from 80-120% (Lesperance et al., 2021). Implementing methods from Klymus et al. (2020), LOD and LOQ were calculated with the cycle threshold (Ct) values from the standard curve in R® (R Core Team, 2024) using the packages *ggplot2* and *drc* (Klymus et al., 2020).

To test eDNA extracted from filters, an assay was created with a total volume of 25µL, including 1µL of extracted eDNA, 12µL of PerfeCTa qPCR ToughMix™, 900 nM of each primer, and 250 nM of internal probe, with the rest of the solution filled with UltraPure™ water. Samples were run with three replicates along with the TaqMan® Exogenous Internal Positive Control (IPC; Applied Biosystems, Waltham, MA) in one of each biological replicate to test for PCR inhibition (Goldberg et al., 2016). The IPC assay and three replicates of no-template controls (NTC) consisted of 5µL of 10x Exo IPC Mix and 1µL of 50x Exo IPC DNA. Additionally, three replicates of no-amplification controls (NAC) were included, which contained 5µL of 10x Exo IPC Mix, 1µL of 50x Exo IPC DNA, and 5µL 10X Exo IPC Block. Each plate included a standard curve including three replicates of each dilution representing 10^5-10^2 copies/µL. Thermocycling conditions were as described above for estimating LOD and LOQ

A cycle threshold (Ct) value represented the qPCR cycle at which the sample has enough DNA to be detected. The Ct value was set to 40 using criteria from Caraguel et. al (2011), due to

the possibility of a false positive from fluorescence breaking down. A positive amplification was defined as two out of the three wells amplifying above a Ct threshold value of 40. If only one replicate was amplified, then the sample was rerun to confirm. A subset of positive qPCR products was sequenced to ensure the presence of the correct amplicon.

To ensure the qPCR amplified the correct product, a PCR was run on two samples at 95°C for 10 minutes, followed by 20 cycles of 95°C for 15 seconds, and 62°C for 1 minute. The PCR product was then visualized using gel electrophoresis as described above. The two PCR products were purified using a standard polyethylene glycol (PEG) cleanup using 25µL of PEG and 25µL of PRC product. The PEG and PCR products were centrifuged at 4°C for 15 minutes. The supernatants were then discarded and 200µL of EtOH stored at -20°C was added to the DNA precipitate. The products were then centrifuged at 4°C for 7 minutes, and supernatants removed. The precipitate was then dried in a speed vacuum with heat until no liquid was visible. Twenty microliters of UltraPure™ were added to the remaining DNA precipitate and were sent off and sequenced by Plasmidsaurus using Oxford Nanopore Technology. A National Center for Biotechnology Information Basic Local Alignment Search Tool (NCBI BLAST) was conducted on the resulting sequences to ensure *E. lachneri* was amplified.

3.2.4 Statistical Analysis

A Logistic Regression model was conducted using the function *glm* in the package *stats* (R Core Team, 2024) to test whether the probability of detecting *E. lachneri* using eDNA is influenced by its abundance at a given site (based on CPUE from Chapter 1). By comparing CPUE and eDNA detection, the amount of *E. lachneri* that were caught could be correlated with the ability to detect *E. lachneri* through eDNA.

To evaluate if the number of *E. lachneri* caught at each site affected the amplification of qPCR, each site's eDNA lowest Ct value and associated CPUE of *E. lachneri* from Chapter I were compared using a Linear Regression model with the function *lm* in the package *stats* (R Core Team, 2024). The CPUE of *E. lachneri* calculated how many individuals were present at a site. With these analyses, the outcome will indicate if a greater CPUE value correlates with having higher number of DNA copies and eDNA detections.

3.3 Results

The complete mitogenome is 16,598 base pairs in length and consists of 13 protein-coding genes, 22 transfer ribonucleic acid (tRNA), 2 ribosomal ribonucleic acid (rRNA), and one D-loop control region. The *E. lachneri* mitogenome was uploaded into NCBI with the accession number PP915961 (Figure 3.1).

Estimates of assay limits were as follows: the LOD was >10 copies, and LOQ was 4,040 with an r^2 value of 0.997. Multi-species amplification tests showed that the assay can amplify DNA from the intended targets *E. lachneri* and *E. ramseyi*. Furthermore, species that co-occur with *E. lachneri* did not amplify.

Of the 39 sites where eDNA was collected, *E. lachneri* DNA was detected in 10 (Table 3.4; Figure 3.2). The PCR inhibition was not suspected in any eDNA samples as the IPC performed as expected. Both samples qPCR products that were sequenced showed a 100% identity to *E. lachneri* (GenBank ID: PP915961). The Logistic Correlation Model indicated that there was a was no correlation between CPUE and eDNA detection (P = 0.36, Z = 0.915, Standard Error = 0.027). A Linear Regression model indicated that there was no correlation between the number of DNA copies detected at each site and CPUE (R^2=0.14, P=0.08, T=1.83).

3.4 Discussion

Validation of the qPCR assay designed to detect members of *Ulocentra* yielded results that indicate it is an assay capable of amplifying low amounts of *E. lachneri* DNA. This is potentially useful given the low abundance of *E. lachneri* at some sites sampled for this study. After sequencing positive qPCR products and matching *E. lachneri* sequences, the positive eDNA detection of the target species has been validated. Future studies of this subgenus can be surveyed with these non-invasive eDNA protocols (Strickland & Roberts, 2019; Takahara et al., 2020). This assay can be used for future projects concerning *E. lachneri* or other snubnose darters including: *Etheostoma scotti*, *Etheostoma raneyi*, *Etheostoma faulkneri*, and *Etheostoma brevirostrum*. Additionally, this assay can be used in conservation projects for species that are endangered or threatened such as *Etheostoma chermocki* and *Etheostoma scotti* (IUCN, 2025). With our current understanding of species distributions and available sequence data, this assay should be applicable in any geographic area where *Ulocentra* spp. are present.

The qPCR assay did not detect *E. lachneri* at sites where physical specimens were not collected. There were also sites where *E. lachneri* was physically detected (Chapter I), but there was no eDNA detection. This could be due to not enough DNA in the water, needing more replicates of each eDNA extraction, needing to filter more water, or needing more DNA template for the assay. During the survey at KMF23081101, an eDNA sample was not collected due to a pump malfunction; thus, there is no eDNA detection data for that site, and it was not included in the analysis. Comparing the traditional survey results with results from eDNA detections, extracted eDNA may need to have more replicates in order for better detection.

Statistical analysis indicated that detection and Ct value of *E. lachneri* in eDNA cannot be predicted based on CPUE. This means that there was no relationship between sites that had a

greater CPUE and whether the site had an eDNA amplicon or eDNA detection, which disagrees with other findings such as Lacoursière-Roussel et al. (2016). It is expected to find greater Ct and detection values at sites with a greater CPUE since it would mean there are more *E. lachneri* individuals releasing DNA into the water and making qPCR detections more likely. While our study suggests that CPUE is not a good indicator of eDNA detection nor Ct value, other variables such as water temperature and pH that were not looked at in this study could be better indicators (Wang et al., 2021).

With a high LOQ value of 4,040, the primer assay may have difficulty assessing the quantity of low-abundance DNA. With a high LOQ value, this may allude to why some traditional surveys had *E. lachneri* present at the site, yet *E. lachneri* was not detected through eDNA. When using eDNA, it is difficult to quantify the number of specimens detected because the results can be altered by the proximity of the fish to the filter, stress of the fish, and water pH (Beng & Corlett, 2020). Due to this, the exact number of *E. lachneri* individuals cannot be determined from qPCR results alone. Other published studies have incorporated known fish biomass with eDNA detections to estimate the abundance of target species (Yates et al., 2025). With new eDNA detection methods, it will be easier to conduct routine surveys of the Tombigbee Darter and determine which streams they inhabit.

3.5 Tables

Table 1.9 Results of qPCR detection of *Etheostoma lachneri* DNA from eDNA at 39 survey sites.

Sample Number	Site	Lat	Long	Site Data	CT-1	CT-2	CT-3	eDNA Y/N	CPUE
10424	DLP23081901	32.5298	-88.512	Ponta Creek	41.74	40.47	43.83	N	8
10417	DLP23081902	32.3659	-88.415	Alamuchee Creek	37.63	37.89	41.43	Y	14
10409	DLP23082601	33.868	-88.569	Wolf Creek	38.01	36.87	39.85	Y	53
10347	DLP23082602	33.6653	-88.28	Mayhew Creek	No Ct	No Ct	No Ct	N	8
10438	KMF23071101	33.8304	-88.944	Chico Creek	No Ct	No Ct	No Ct	N	0
10429	KMF23071102	34.087	-88.991	Redland Creek	No Ct	No Ct	No Ct	N	22
10435	KMF23071801	34.5194	-88.446	Hurricane Creek	39.82	No Ct	No Ct	N	2
10434	KMF23071802	34.6677	-88.429	Big Brown Creek	No Ct	No Ct	No Ct	N	0
10428	KMF23072601	33.5088	-88.393	Luxapalila Creek	38.99	No Ct	No Ct	N	0

Table 3.1 (continued)

Sample Number	Site	Lat	Long	Site Data	CT-1	CT-2	CT-3	eDNA Y/N	CPUE
10406	KMF23072602	33.4776	-88.363	McCary Creek	43.66	41.99	No Ct	N	0
10412	KMF23072603	33.48	-88.36	Catalpa-Tibbee Creek	40.73	No Ct	No Ct	N	0
10440	KMF23072801	33.0447	-88.817	Moody Creek	No Ct	No Ct	No Ct	N	1
10425	KMF23072802	32.8356	-88.659	Straight Creek	40	42.86	No Ct	N	7
10433	KMF23073101	33.8323	-88.475	Nichols Creek	No Ct	No Ct	No Ct	N	2
10420	KMF23073102	33.8559	-88.319	unnamed Creek-Buttahatchee River	39.74	44.32	44.32	N	0
10423	KMF23073103	33.7577	-88.418	Wilson Creek	39.32	40.66	No Ct	N	8
10432	KMF23080301	34.0899	-88.35	Bluff Creek	38.9	38.63	No Ct	Y	5
10418	KMF23080302	34.1185	-88.316	Smith Creek	39.16	40.114	43.93	N	5
10421	KMF23080303	34.2303	-88.258	Johns Creek	36.65	36.972	41.30	Y	44

Table 3.1 (continued)

Sample Number	Site	Lat	Long	Site Data	CT-1	CT-2	CT-3	eDNA Y/N	CPUE
10416	KMF23081101	32.6896	-88.616	Pawticfaw Creek	No Ct	No Ct	No Ct	N	16
10422	KMF23081401	34.2859	-88.219	Briar Creek	No Ct	No Ct	No Ct	N	27
10414	KMF23081402	34.385	-88.206	Gum Creek	37	42.89	44.08	N	33
10427	KMF23081403	34.3045	-88.355	Dulaney Branch	37.515	36.811	No Ct	Y	4
10408	KMF23081501	34.4718	-88.261	Red Bud Creek	36.7	37.35	37.83	Y	8
10426	KMF23081502	34.6682	-88.285	Hurricane Creek	No Ct	No Ct	No Ct	N	0
10419	KMF23090401	33.9441	-88.488	Weaver Creek	38.799	38.693	42.43	Y	15
10407	KMF23090402	34.1636	-88.511	Boguefala Creek	38.51	38.93	43.20	Y	11
10410	KMF23090403	34.2297	-88.667	Tulip Creek	34.01	31.85	No Ct	Y	14
10411	KMF23090901	34.5837	-88.628	Twenty Mile Creek	39.78	42.7	42.80	N	0
10348	KMF23090902	34.3093	-88.51	Mantachie Creek	No Ct	No Ct	No Ct	N	18

Table 3.1 (continued)

Sample Number	Site	Lat	Long	Site Data	CT-1	CT-2	CT-3	eDNA Y/N	CPUE
10415	KMF23090903	34.4847	-88.506	Donivan Creek	No Ct	No Ct	No Ct	N	6
10405	KMF23091301	33.3822	-88.313	Nash Creek	34.52	34.72	40.55	Y	3
10436	KMF23091302	33.4094	-88.334	Ellis Creek	No Ct	No Ct	No Ct	N	5
10437	KMF23091303	33.5353	-88.304	Brush Creek	No Ct	No Ct	No Ct	N	3
10345	KMF23092301	34.1099	-88.722	Chiwapa Creek	No Ct	No Ct	No Ct	N	0
10346	KMF23092302	34.2208	-88.999	Webster Creek	37.75	No Ct	No Ct	N	0
10439	MSW23062301	32.9938	-88.655	Running Water Creek	No Ct	No Ct	No Ct	N	0
10430	MWS23062302	33.1223	-88.806	Sun Creek	No Ct	No Ct	No Ct	N	0
10431	MWS23062701	33.1975	-88.995	Mill Creek	No Ct	No Ct	No Ct	N	9

Abbreviations: Catch per Unit Effort (CPUE), cycle threshold (Ct), Y was used to represent a positive amplicon of *E. lachneri* in the extracted eDNA sample, N was used to represent a negative amplicon of *E. lachneri* in the extracted eDNA sample.

Table 1.10 *Ulocentra* specific primers were created using *E. lachneri* mitogenome (accession number PP915961) to amplify *E. lachneri* in extracted eDNA filters.

ND6 Primer	**Sequence**
Forward Primer	gctttgctggcctctattttagtac
Reverse Primer	atcctagggctttgttctctagc
Internal Oligo	cccctctcgcaattcctgttctgaa

3.6 Figures

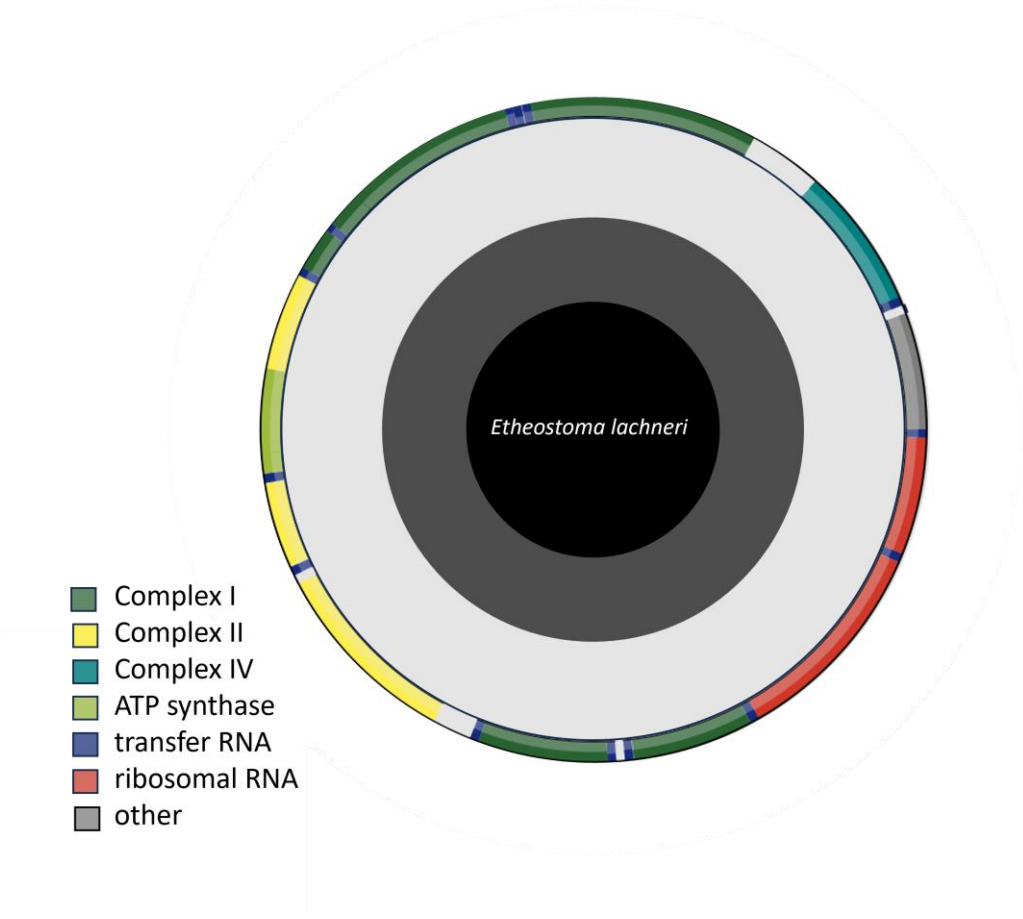

Figure 3.1 Annotated mitochondrial genome for *Etheostoma lachneri.*

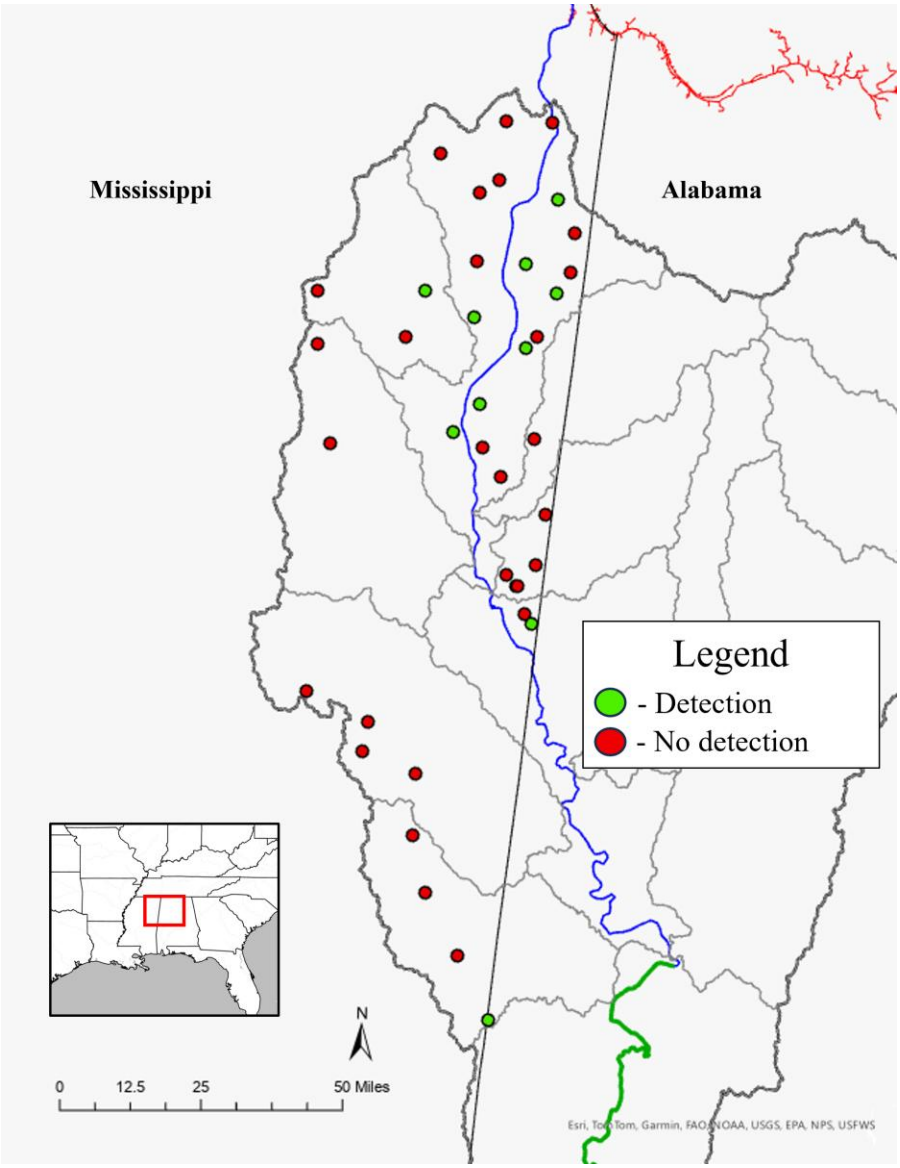

Figure 1.13 Map of eDNA survey sites in Northeast Mississippi, with colored dots representing if *E. lachneri* were detected.

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APPENDIX A

SUPPLEMENTAL TABLE FOR CHAPTER I

Table A.1 Total species observations resulting from 40 standardized aquatic faunal surveys.

Family	**Genus**	**Species**	**Total**
Aphredoderidae	*Aphredoderus*	*sayanus*	3
Astacidae	*Crayfish*	spp.	174
Atherinidae	*Labidesthes*	*vanhyningi*	4
Cambaridae	*Cambarus*	*striatus*	5
Catostomidae	*Carpiodes*	*cyprinus*	0
Catostomidae	*Erimyzon*	*claviformes*	3
Catostomidae	*Erimyzon*	*oblongus*	20
Catostomidae	*Hypentelium*	*etowanum*	63
Catostomidae	*Ictalurus*	*bubalus*	1
Catostomidae	*Minytrema*	*melanops*	3
Catostomidae	*Moxostoma*	*poecilurum*	37
Catostomidae	*Moxostoma*	spp.	1
Centrarchidae	*Ambloplites*	*ariommus*	4
Centrarchidae	*Elassoma*	*zonatum*	11
Centrarchidae	*Lepomis*	*cyanellus*	153
Centrarchidae	*Lepomis*	*gulosus*	7
Centrarchidae	*Lepomis*	*macrochirus*	312
Centrarchidae	*Lepomis*	*marginatus*	1
Centrarchidae	*Lepomis*	*megalotis*	144
Centrarchidae	*Lepomis*	*microlophus*	14
Centrarchidae	*Lepomis*	*miniatus*	4
Centrarchidae	*Lepomis*	*punctatus*	1
Centrarchidae	*Lepomis*	*solis*	433
Centrarchidae	*Micropterus*	*henshalli*	66
Centrarchidae	*Micropterus*	*salmoides*	12
Centrarchidae	*Micropterus*	spp.	1
Corydalidae	*Hellgrammite*	spp.	4
Cyprinidae	*Campostoma*	*anomalum*	520
Cyprinidae	*Cyprinella*	*venusta*	843
Cyprinidae	*Hybognathus*	*nuchalis*	48
Cyprinidae	*Hybopsis*	*wincelli*	16
Cyprinidae	*Luxilus*	*chrysocephalus*	603
Cyprinidae	*Lythrurus*	*bellas*	408
Cyprinidae	*Macrhybopsis*	spp.	10
Cyprinidae	*Nocomis*	*leptocephalus*	238
Cyprinidae	*Notemigonus*	*crysoleucas*	0

Table A.1 (continued)

Family	**Genus**	**Species**	**Total**
Cyprinidae	*Notropis*	*ammophilus*	770
Cyprinidae	*Notropis*	*atherinoides*	1
Cyprinidae	*Notropis*	*baileyi*	1017
Cyprinidae	*Notropis*	*buccata*	9
Cyprinidae	*Notropis*	*candidus*	2
Cyprinidae	*Notropis*	spp.	9
Cyprinidae	*Notropis*	*texanus*	132
Cyprinidae	*Notropis*	*volucellus*	152
Cyprinidae	*Pimephales*	*notatus*	922
Cyprinidae	*Pimephales*	*vigilax*	69
Cyprinidae	*Semotilus*	*atromaculatus*	268
Cyprinidae	*Semotilus*	*thoreauianus*	9
Cyrenidae	*Corbicula*	spp.	1
Ephemerellidae	*Mayfly*	spp.	19
Esocidae	*Esox*	*americanus*	3
Fundulidae	*Fundulus*	*notatus*	58
Fundulidae	*Fundulus*	*olivaceous*	95
Ictaluridae	*Amieurus*	*natalis*	4
Ictaluridae	*Ictiobus*	*punctatus*	11
Ictaluridae	*Noturus*	*funebris*	1
Ictaluridae	*Noturus*	*gyrinus*	19
Ictaluridae	*Noturus*	*leptacanthus*	112
Ictaluridae	*Noturus*	*munitus*	12
Ictaluridae	*Noturus*	*nocturnus*	58
Ictaluridae	*Noturus*	*phaeus*	2
Palaemonidae	*Palaemonetes*	spp.	2
Percidae	*Ammocrypta*	*beanii*	3
Percidae	*Ammocrypta*	*meridiana*	1
Percidae	*Crystallaria*	*asprella*	9
Percidae	*Etheostoma*	*artesiae*	76
Percidae	*Etheostoma*	*chlorosomum*	10
Percidae	*Etheostoma*	*fusiforme*	1
Percidae	*Etheostoma*	*lachneri*	384
Percidae	*Etheostoma*	*nigrum*	143
Percidae	*Etheostoma*	*parvipinne*	72
Percidae	*Etheostoma*	*proeliare*	13
Percidae	*Etheostoma*	*rupestre*	187
Percidae	*Etheostoma*	spp.	8
Percidae	*Etheostoma*	*stigmaeum*	419

Table A.1 (continued)

Family	**Genus**	**Species**	**Total**
Percidae	*Etheostoma*	*swainii*	47
Percidae	*Percina*	*kathae*	3
Percidae	*Percina*	*maculata*	2
Percidae	*Percina*	*nigrofasciata*	336
Percidae	*Percina*	*sciera*	32
Percidae	*Percina*	spp.	2
Percidae	*Percina*	*suttkusi*	1
Percidae	*Percina*	*vigil*	28
Petromyzontidae	*Ichthyomyzon*	*gagei*	0
Petromyzontidae	*Lamprey*	spp.	85
Plethodontidae	*Salamander*	spp.	11
Poeciliidae	*Gambusia*	*affinis*	935
Ranidae	*Tadpole*	spp.	26
		Total	10758

APPENDIX B

SUPPLEMENTAL TABLES AND FIGURE FOR CHAPTER II

Table B.1 *Etheostoma lachneri* individuals sequenced for mitochondrial cytochrome *b* gene using Psomagen and their locality.

ID	Site ID	Lat	Long	Population	GenID
9995	KMF2307180202	34.6678	-88.429	Everywhere else	PV595394
9976	KMF2307110203	34.087	-88.991	Everywhere else	PV595395
10038	KMF2307310101	33.8323	-88.475	Everywhere else	PV595396
10069	KMF2307280202	32.8356	-88.659	Sucarnoochee	PV595397
10072	KMF2307280205	32.8356	-88.659	Sucarnoochee	PV595398
ml01	ml	34.294	-88.614	Everywhere else	PV595399
ml02	ml	34.294	-88.614	Everywhere else	PV595400
gx01	gx	34.1777	-88.954	Everywhere else	PV595401
gx02	gx	34.1777	-88.954	Everywhere else	PV595402
we02	we	33.9675	-88.395	Everywhere else	PV595403
we01	we	33.9675	-88.395	Everywhere else	PV595404
Gr01	gr	33.7399	-88.386	Everywhere else	PV595405
gr02	gr	33.7399	-88.386	Everywhere else	PV595406
bu204	bu	34.1701	-87.981	Everywhere else	PV595407
ch02	ch	34.0603	-88.959	Everywhere else	PV595408
ss01	ss	33.9429	-88.97	Everywhere else	PV595409
rk01	rk	34.5191	-88.257	Everywhere else	PV595410
rk02	rk	34.5191	-88.257	Everywhere else	PV595411
tb01	tb	34.6456	-88.357	Everywhere else	PV595412
tb02	tb	34.6456	-88.357	Everywhere else	PV595413
ts01	ts	32.9487	-87.962	Everywhere else	PV595414
ts02	ts	32.9487	-87.962	Everywhere else	PV595415
Km01	km	32.4753	-88.067	Everywhere else	PV595416
km02	km	32.4753	-88.067	Everywhere else	PV595417
pk01	pk	32.4833	-88.102	Everywhere else	PV595418
pk02	pk	32.4833	-88.102	Everywhere else	PV595419
hr01	hr	32.5496	-87.953	Everywhere else	PV595420
hr02	hr	32.5496	-87.953	Everywhere else	PV595421
by01	by	31.9142	-87.804	Everywhere else	PV595422
by02	by	31.9142	-87.804	Everywhere else	PV595423
su202	su	32.3432	-88.449	Everywhere else	PV595424
nw202	nw	34.069	-87.664	Everywhere else	PV595425

Table B.1 (continued)

ID	Site ID	Lat	Long	Population	GenID
nw302	nw	34.018	-87.718	Everywhere else	PV595426
su104	su	32.68	-88.649	Sucarnoochee	PV595427
su201	su	32.3432	-88.449	Everywhere else	PV595428
be01	be	31.9875	-88.331	Everywhere else	PV595429
be02	be	31.9875	-88.331	Everywhere else	PV595430
bu205	bu	34.1701	-87.981	Everywhere else	PV595431
bu101	bu	34.1764	-87.789	Everywhere else	PV595432
bu102	bu	34.1764	-87.789	Everywhere else	PV595433
bu202	bu	34.1701	-87.981	Everywhere else	PV595434
bu301	bu	34.1122	-87.731	Everywhere else	PV595435
bu302	bu	34.1122	-87.731	Everywhere else	PV595436
ccr01	ccr	33.6008	-88.308	Everywhere else	PV595437
ccr02	ccr	33.6008	-88.308	Everywhere else	PV595438
cw102	cw	30.922	-88.193	Mobile-Tensaw Delta	PV595439
cw201	cw	30.949	-88.224	Mobile-Tensaw Delta	PV595440
cw202	cw	30.949	-88.224	Mobile-Tensaw Delta	PV595441
gg01	gg	33.3493	-87.926	Everywhere else	PV595442
gg02	gg	33.3493	-87.926	Everywhere else	PV595443
ub01	ub	32.2154	-87.942	Everywhere else	PV595444
ub02	ub	32.2154	-87.942	Everywhere else	PV595445
er01	er	31.119	-88.165	Everywhere else	PV595446
scr01	scr	33.3761	-89.109	Everywhere else	PV595447
mm02	mm	33.2748	-87.632	Lower Black Warrior	PV595448
mr01	mr	32.8809	-87.86	Lower Black Warrior	PV595449
mr02	mr	32.8809	-87.86	Lower Black Warrior	PV595450
mr03	mr	32.8809	-87.86	Lower Black Warrior	PV595451
mr04	mr	32.8809	-87.86	Lower Black Warrior	PV595452
mm01	mm	33.2748	-87.632	Lower Black Warrior	PV595453

Table B.1 (continued)

ID	Site ID	Lat	Long	Population	GenID
el02	el	32.9543	-87.479	Lower Black Warrior	PV595454
su 302	su	32.5493	-88.497	Sucarnoochee	PV595455
su101	su	32.68	-88.649	Sucarnoochee	PV595456
SU103	su	32.68	-88.649	Sucarnoochee	PV595457
su401	su	32.8355	-88.66	Sucarnoochee	PV595458
su102	su	32.68	-88.649	Sucarnoochee	PV595459
SU303	su	32.5493	-88.497	Sucarnoochee	PV595460
SU402	su	32.8355	-88.66	Sucarnoochee	PV595461
su203	su	32.3432	-88.449	Everywhere else	PV595462
su301	su	32.5493	-88.497	Sucarnoochee	PV595463

Table B.2 *Etheostoma lachneri** individuals sequenced for nuclear DNA using DArTseq™ and their locality.

ID	Site ID	Lat	Long	Stream	Population
TM-3	TM	34.54561	-88.6189	Twenty Mile Creek	Upper_Tombigbee
BM01	BM	34.20821	-88.257	U.T. Bull Mountain Creek	Upper_Tombigbee
10042	KMF23073103	33.75773	-88.4184	Tombigbee R, Mckinley Creek, Wilson Creek	Upper_Tombigbee
10038	KMF23073101	33.83232	-88.4749	Tombigbee R, Mckinley Creek, Nichols Creek @ DanTaylor rd	Upper_Tombigbee
10284	KMF23090902	34.3093	-88.5101	Mantaohie Creek	Upper_Tombigbee

Table B.2 (continued)

ID	Site ID	Lat	Long	Stream	Population
10274	KMF23081501	34.47184	-88.2609	Mackeys Creek	Upper_Tombigbee
9995	KMF23071802	34.66778	-88.4294	Big Brown-Tomigbee	Upper_Tombigbee
ml01	ML	34.294	-88.614	Tulip Creek	Upper_Tombigbee
ml02	ML	34.294	-88.614	Tulip Creek	Upper_Tombigbee
ml03	ML	34.294	-88.614	Tulip Creek	Upper_Tombigbee
ml04	ML	34.294	-88.614	Tulip Creek	Upper_Tombigbee
gx01	GX	34.17769	-88.9536	Garrett Creek	Upper_Tombigbee
gx02	GX	34.17769	-88.9536	Garrett Creek	Upper_Tombigbee
gx03	GX	34.17769	-88.9536	Garrett Creek	Upper_Tombigbee
gx04	GX	34.17769	-88.9536	Garrett Creek	Upper_Tombigbee
ch02	CH	34.06034	-88.9589	Mill Creek	Upper_Tombigbee
9976	KMF23071102	34.08701	-88.9913	Chipwa Creek	Upper_Tombigbee
CH01	CH	34.06034	-88.9589	Mill Creek	Upper_Tombigbee
CH07	CH	34.06034	-88.9589	Mill Creek	Upper_Tombigbee
9975	KMF23071102	34.08701	-88.9913	Chipwa Creek	Upper_Tombigbee
9977	KMF23071102	34.08701	-88.9913	Chipwa Creek	Upper_Tombigbee
bu301	BU3	34.11216	-87.7314	Buttahatchie River	Upper_Tombigbee
bu101	BU1	34.17638	-87.789	Barn Creek	Upper_Tombigbee
bu204	BU2	34.17006	-87.9809	Williams Creek	Upper_Tombigbee
mmns5949	MMNS	33.94897	-88.2543	Sipsey Creek	Upper_Tombigbee
bu102	BU1	34.17638	-87.789	Barn Creek	Upper_Tombigbee
mmns5950	MMNS	33.94897	-88.2543	Sipsey Creek	Upper_Tombigbee
ccr01	CCR	33.60077	-88.3075	Cooper Creek	Upper_Tombigbee
10385	DLP23082602	33.66532	-88.2796	Lower Yellow Creek	Upper_Tombigbee

Table B.2 (continued)

ID	Site ID	Lat	Long	Stream	Population
ccr02	CCR	33.60077	-88.3075	Cooper Creek	Upper_Tombigbee
10387	DLP23082602	33.66532	-88.2796	Lower Yellow Creek	Upper_Tombigbee
10386	DLP23082602	33.66532	-88.2796	Lower Yellow Creek	Upper_Tombigbee
10388	DLP23082602	33.66532	-88.2796	Lower Yellow Creek	Upper_Tombigbee
10389	DLP23082602	33.66532	-88.2796	Lower Yellow Creek	Upper_Tombigbee
nw302	NW3	34.018	-87.718	UT Little New River	Upper_Tombigbee
nw202	NW2	34.069	-87.664	Crooked Creek	Upper_Tombigbee
nw201	NW2	34.069	-87.664	Crooked Creek	Upper_Tombigbee
nw301	NW3	34.018	-87.718	UT Little New River	Upper_Tombigbee
nw303	NW3	34.018	-87.718	UT Little New River	Upper_Tombigbee
9990	MWS23062701	33.1975	-88.9947	Little Noxubee River-Noxubee River	Upper_Tombigbee
LN01	LN	33.17883	-89.0551	Little Noxubee River	Upper_Tombigbee
scr01	scr	33.37607	-89.1088	Sand Creek	Upper_Tombigbee
9993	MWS23062701	33.1975	-88.9947	Little Noxubee River-Noxubee River	Upper_Tombigbee
scr02	scr	33.37607	-89.1088	Sand Creek	Upper_Tombigbee
scr03	scr	33.37607	-89.1088	Sand Creek	Upper_Tombigbee

Table B.2 (continued)

ID	Site ID	Lat	Long	Stream	Population
Ln02	LN	33.17883	-89.0551	Little Noxubee River	Upper_Tombigbee
Ln03	LN	33.17883	-89.0551	Little Noxubee River	Upper_Tombigbee
ts01	TS	32.94874	-87.9616	Trussels Creek	Upper_Tombigbee
GG04	GG	33.34927	-87.9256	Hog Branch	Upper_Tombigbee
10294	KMF23091302	33.40947	-88.3336	Kincaide Creek, Tombigbee Rvr, Ellis Creek	Upper_Tombigbee
10391	KMF23091301	33.38222	-88.3131	Kincaide Creek, Tombigbee Rvr, Nash Creek	Upper_Tombigbee
GG05	GG	33.34927	-87.9256	Hog Branch	Upper_Tombigbee
ts02	TS	32.94874	-87.9616	Trussels Creek	Upper_Tombigbee
10390	KMF23091301	33.38222	-88.3131	Kincaide Creek, Tombigbee Rvr, Nash Creek	Upper_Tombigbee
10295	KMF23091302	33.40947	-88.3336	Kincaide Creek, Tombigbee Rvr, Ellis Creek	Upper_Tombigbee
mr01	MR	32.88086	-87.8598	Minter Creek	Lower_Black_Warrior
el02	EL	32.95427	-87.479	Elliot's Creek	Lower_Black_Warrior
sd01	SD	33.05981	-87.448	Bunch Creek (Big Sandy)	Lower_Black_Warrior
old-mm02	MM	33.27483	-87.6316	Mill Creek	Lower_Black_Warrior
mr04	MR	32.88086	-87.8598	Minter Creek	Lower_Black_Warrior

Table B.2 (continued)

ID	Site ID	Lat	Long	Stream	Population
old-mm01	MM	33.27483	-87.6316	Mill Creek	Lower_Black_Warrior
sd02	SD	33.05981	-87.448	Bunch Creek (Big Sandy)	Lower_Black_Warrior
EL03	EL	32.95427	-87.479	Elliot's Creek	Lower_Black_Warrior
10213	KMF23081101	32.68968	-88.6165	Pawticfaw Creek	Lower_Tombigbee
10268	DLP23081902	32.3659	-88.4153	Alamunchee Creek	Lower_Tombigbee
km01	KM	32.4753	-88.0668	Mill Creek	Lower_Tombigbee
10223	KMF23081102	32.62234	-88.6213	Pawticfaw Creek	Lower_Tombigbee
su201	SU2	32.34323	-88.4486	Alamuchee Creek	Lower_Tombigbee
pk02	PK	32.48331	-88.1022	Ponkabia Creek	Lower_Tombigbee
10263	DLP23081901	32.52988	-88.5119	Big Reed Creek-Ponta Creek	Lower_Tombigbee
10072	KMF23072801	33.04477	-88.8173	Hashuqua	Lower_Tombigbee
BY04	BY	31.91417	-87.8044	Bashi Creek	Lower_Tombigbee
be01	BE	31.98749	-88.3313	Bougeloosa Creek	Lower_Tombigbee
UB03	UB	32.21535	-87.9415	Little Beaver Creek	Lower_Tombigbee
hr01	HR	32.54957	-87.9527	Hall Creek	Lower_Tombigbee
hr02	HR	32.54957	-87.9527	Hall Creek	Lower_Tombigbee
ub02	UB	32.21535	-87.9415	Little Beaver Creek	Lower_Tombigbee
by01	BY	31.91417	-87.8044	Bashi Creek	Lower_Tombigbee
be02	BE	31.98749	-88.3313	Bougeloosa Creek	Lower_Tombigbee
old-cw102	CW1	30.922	-88.193	Chickasaw Creek	Lower_Tombigbee
cw201	CW2	30.94904	-88.2235	UT Chickasaw Creek	Lower_Tombigbee
cw202	CW2	30.94904	-88.2235	UT Chickasaw Creek	Lower_Tombigbee

Table B.2 (continued)

ID	Site ID	Lat	Long	Stream	Population
er02	ER	31.119	-88.165	Cedar Creek	Lower_Tombigbee
er03	ER	31.119	-88.165	Cedar Creek	Lower_Tombigbee
old-cw101	CW1	30.922	-88.193	Chickasaw Creek	Lower_Tombigbee
10587	CNT24032901	31.78678	-87.7182	Bassett Creek	Lower_Tombigbee
10588	CNT24032901	31.78678	-87.7182	Bassett Creek	Lower_Tombigbee
10589	CNT24032901	31.78678	-87.7182	Bassett Creek	Lower_Tombigbee
10590	CNT24032901	31.78678	-87.7182	Bassett Creek	Lower_Tombigbee
10591	CNT24032901	31.78678	-87.7182	Bassett Creek	Lower_Tombigbee
10592	CNT24032901	31.78678	-87.7182	Bassett Creek	Lower_Tombigbee
10593	CNT24032902	31.61972	-87.8422	Rabbit Creek	Lower_Tombigbee
10660	CNT24050901	33.25657	-87.0786	Mud Creek, Alabama	*E. ramseyi*
10662	CNT24050901	33.25657	-87.0786	Mud Creek, Alabama	*E. ramseyi*

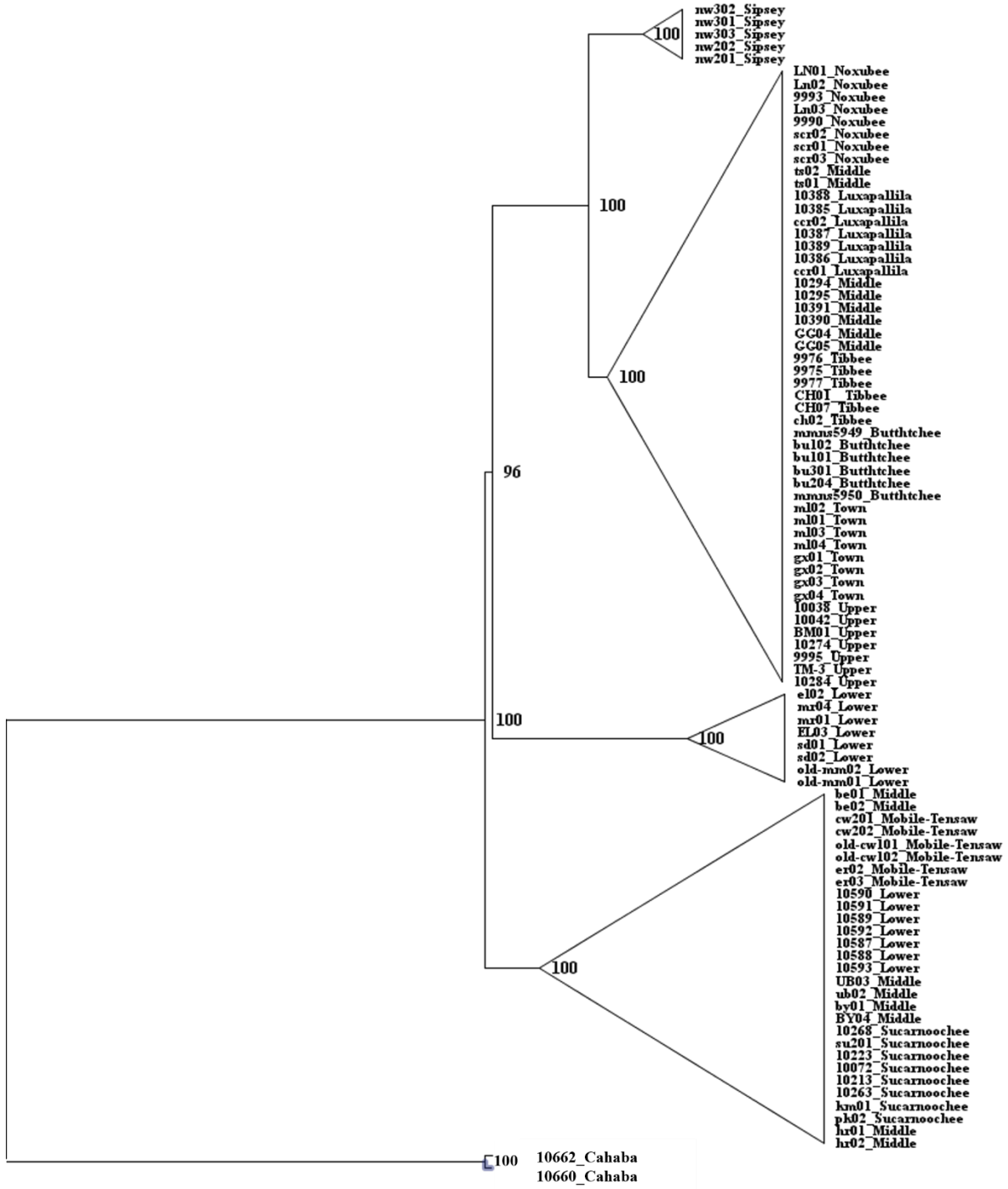

Figure B.1 Maximum-likelihood phylogenetic tree (substitution model: TVM+F+G4) for *E. lachneri* using 92 single-nucleotide polymorphism sequencing with 2 *E. ramseyi* (Cahaba) used as an outgroup and bootstrap support values present.

APPENDIX C

SUPPLEMENTAL TABLE FOR CHAPTER III

Table A.1 Example qPCR plate layout

	1	2	3	4	5	6	7	8	9	10	11	12
A	10405-1	10407-3	10410-2	10413-1	10415-3	10418-2	10421-1	10423-3	10426-2	10429-1	NAC	10^4
B	10405-2	10408-1	10410-3	10413-2	10416-1	10418-3	10421-2	10424-1	10426-3	10429-2	NTC	10^4
C	10405-3	10408-2	10411-1	10413-3	10416-2	10419-1	10421-3	10424-2	10427-1	10429-3	NTC	10^3
D	10406-1	10408-3	10411-2	10414-1	10416-3	10419-2	10422-1	10424-3	10427-2	10430-1	NTC	10^3
E	10406-2	10409-1	10411-3	10414-2	10417-1	10419-3	10422-2	10425-1	10427-3	10430-2	10^5	10^3
F	10406-3	10409-2	10412-1	10414-3	10417-2	10420-1	10422-3	10425-2	10428-1	10430-3	10^5	10^2
G	10407-1	10409-3	10412-2	10415-1	10417-3	10420-2	10423-1	10425-3	10428-2	NAC	10^5	10^2
H	10407-2	10410-1	10412-3	14015-2	10418-1	10420-3	10423-2	10426-1	10428-3	NAC	10^4	10^2

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